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CURRICULUM VITAE

NAME: Stephen D. Klyce, Ph.D., FARVO

BIRTH: October 30, 1942 - Arlington, Massachusetts

EDUCATION

1964 B.S. (Zoology), University of Massachusetts

1971 Ph.D. (Physiology), Yale University

POSITIONS HELD

1964-1967 Research Assistant, Department of Cornea Research, Retina Foundation, Boston, Massachusetts

1967-1971 Research Assistant, Graduate Student, Department of Physiology, Yale University Graduate School, New Haven, Connecticut

1971-1972 Research Associate, Department of Ophthalmology and the Visual Sciences, Yale University School of Medicine, New Haven, Connecticut

1972-2008 Principal Investigator, NEI RO1-EY03311, Integrated Assessment of Corneal Form and Function.

1972-1979 Research Associate, Senior Research Associate, Division of Ophthalmology, Department of Surgery, Stanford University School of Medicine, Stanford, California

1979-2008 Professor of Ophthalmology, Department of Ophthalmology, Louisiana State University Health Sciences Center, New Orleans, Louisiana

1987-2008 Graduate Faculty Member, Medical Center Graduate Council, Louisiana State University

1990-1994 Director, LSU Training Program in Vision Research

1990-2008 Adjunct Professor of Biomedical Engineering, Tulane University, New Orleans, Louisiana

1992-2008 Adjunct Professor, Department of Cell Biology and Anatomy, Louisiana State University Health Sciences Center, New Orleans, Louisiana

2008- Adjunct Professor of Ophthalmology, Icahn School of Medicine at Mount Sinai, New York, NY

2008- Ad hoc scientific, clinical trials, and biomedical engineering consultant

FEDERAL/PROFESSIONAL SOCIETY ADVISORY COMMITTEES

1979-	Special reviewer, NIH DRG Visual Sciences A1 Study Section, NIH DRG Special Study Sections, National Science Foundation
1980-1981	Corneal Diseases Panel Member, National Advisory Eye Council
1984-1985	Corneal Diseases Panel Member, National Advisory Eye Council
1984-1992	Ad Hoc Consultant to Prospective Evaluation of Radial Keratotomy (PERK) Study, National Eye Institute
1988	Consultant, National Research Council Committee on Vision on the Use of Contact Lenses under Adverse Conditions
1988-1992	Member, NIH DRG Visual Sciences A1 Study Section
1989-1990	Steering Committee Member, NEI Workshop on Corneal Biophysics
1990	Invited Participant, Meeting on Policy Issues, Vision Research Program Planning Subcommittee, National Advisory Eye Council
1994-1998	Member, Corneal Topography Standards Effort, American National Standards Institute Committee on Ophthalmic Optics (ANSI).
2004-	American Society of Cataract and Refractive Surgery representative to the American National Standards Institute Committee on Ophthalmic Optics (ANSI).
2004-	Committee member, ISO/TC 172/SC 7/WG 7, International Organization for Standardization
2005-	American Society of Cataract and Refractive Surgery FDA Committee
2010-	Member, *Z80.32 - Methodology for Representation of Optically-Induced Visual Phenomena* Effort, American National Standards Institute Committee on Ophthalmic Optics.
2011-2013	Team Leader, Corneal Topography Standard, the Digital Imaging and Communications in Medicine (DICOM) WG-9 workgroup for the American Academy of Ophthalmology.
2018-2020	Chairman, ASCRS•ASOA Telemedicine Task Force
2024-	Member, FDA and Innovation Committee, ASCRS
2025-	Member, Revision of Corneal Topography Map IOD, the Digital Imaging and Communications in Medicine (DICOM) WG-9 workgroup, American Academy of Ophthalmology

PROFESSIONAL RECOGNITION

1976-1980	ARVO Program Committee member, Physiology and Pharmacology Section
1980	ARVO Program Chairman, Southern Section Meeting
1985-1990	ARVO Trustee, Physiology and Pharmacology Section
1988-1989	ARVO Vice President
1990	ARVO President
1986-1990	Member of Council, International Society for Contact Lens Research
1990-1997	Vice President, International Society for Contact Lens Research
1997-1999	President Elect, International Society for Contact Lens Research
1999-2001	President, International Society for Contact Lens Research
2002-	Member of the Executive, International Society for Contact Lens Research
1990-1992	Program Organizer, Cornea & Connective Tissue Section, Xth International Congress of Eye Research
1992-1996	Councilor, International Society for Eye Research
1991-1992	Member, Ophthalmic and Visual Optics Organizing Committee, Optical Soc. of America
1988-1995	Member, Technology Advisory Panel of the LSU Technology Transfer Program
1995-2000	Member, LSUMC-Wide Research Planning and Technology Transfer Committee
1980,1981	Scientific Review Committee, Consultant, Fight for Sight, Inc.
1984-1986	Fellowship Review Committee Member for Fight for Sight, Inc.
1993-2000	Member, Board of Medical Advisors, National Keratoconus Foundation
1993-1996	Member, Corneal Topography Standards Committee, International Society for Refractive Surgery.
1994-1995	Member, Program Committee, Keratoconus Collaborative Conference, National Keratoconus Foundation
2004-2006	American Academy of Ophthalmology Committee on Research Scientist Integration
2004-	American Society of Cataract and Refractive Surgery FDA Committee

2004-	American Society of Cataract and Refractive Surgery ANSI Representative
2004-	Expert, International Organization for Standardization, ISO/TC172 Optics and Optical Instruments, SC7 - Ophthalmic optics and instruments, WG6 Ophthalmic instruments and test methods, WG7 Ophthalmic implants
2010-2016	Scientific Advisory Board, LENSAR, Inc
2011-2016	Board of Directors, Louisiana Eye Research Institute
2012-2014	Lead Coordinator for Ocular Imaging, Scientific Program Committee, 2014 World Ophthalmology Congress (WOC2014), Tokyo, Japan.
2013-2019	Medical Advisory Board, The Eye Bank for Sight Restoration, Inc., Manhattan, NY
2014-2015	Coordinator for Ocular Imaging, Scientific Program Committee, Asia-Pacific Academy of Ophthalmology Congress 2015, Guangzhou, China.
2016-	Member, New York Intra Ocular Lens Implant Society (NYIOLIS)
2017-	Member, Yale Ophthalmological Society
2022-	Member, International Intra-Ocular Implant Club (IIIC)
2022-	International Medical Advisory Board, Optical Express
2023-	Charter Member, World College of Ophthalmology
2023	No. 11 Most cited author in Refractive Surgery. Randleman JB, et al. *J Refract Surg* 2023;39:78-88.
2023	No. 15 and 25 Most cited papers in Refractive Surgery. Randleman JB, et al. *J Refract Surg* 2023;39:78-88.
2024	Member, International Outreach Committee of the Keratoconus & CXL Experts Meeting
2025	The Scholar GPS rankings. Corneal Topography all time #1 Stephen D. Klyce

EDITORIAL BOARDS

1981-1993	Editorial Board, *Current Eye Research*
1984-1990	Editorial Board, *American Journal of Physiology: Cell Physiology*
1987-92 2002-07	

2012-2023	Editorial Board, *Investigative Ophthalmology and Visual Science*
1988-	Editorial Board, *Cornea*
1994-1996	Editorial Board, *Eyecare Technology*
1989	Guest Editor, Nov/Dec issue, *Refractive and Corneal Surgery*
1991-2012	Executive Editor, *Experimental Eye Research*
2000-2005	Editorial Board, *Review of Refractive Surgery*
2001-	Editorial Board, *Journal of Toxicology. Cutaneous and Ocular Toxicology*
2003-	Editorial Board, *Journal of Refractive Surgery*
2004-	Editorial Board, *Ophthalmologia*
2008-	Editorial Board, *Journal of Optometry*
2012-	Editorial Board, *International Journal of Keratoconus and Ectatic Corneal Diseases*
2013-	Editorial Board, *Eye & Contact Lens*

PROFESSIONAL AWARDS

1986	Alcon Research Institute Recognition Award for Ophthalmic Research
1989	Broadhurst Lecturer, Eye Research Institute, Boston
1990	Everett Kinsey Lecturer, Contact Lens Association of Ophthalmology
1991	Lans Lecturer, International Society for Refractive Keratoplasty
1991	Special Recognition Award, ARVO
1991	Max Schapiro Memorial Lecturer, American Academy of Optometry
1991	Best Lecturer, Bausch & Lomb European Research Symposium
1992	Honor Award, American Academy of Ophthalmology
1993	Holland Guest Lecture, European Contact Lens Association of Ophthalmologists

1996 Best Session Paper, III International Meeting on Cataract and Refractive Surgery, Belo Horizonte, Brazil

1996 Whitney Sampson Lecture, Contact Lens Association of Ophthalmology and American Academy of Ophthalmology

1996 University of Houston Distinguished Researcher Award

2000 Charles Kelman Innovator Award, American Society of Cataract and Refractive Surgery

2001 Innovator Award, Pan Arab African Congress of Ophthalmology

2003 The Ruben Medal, International Society for Contact Lens Research

2005 Senior Achievement Award, American Academy of Ophthalmology

2007 Casebeer Lecture Award, International Society for Refractive Surgery of the American Academy of Ophthalmology

2008 Research & Innovation Award, Italian Refractive Surgery Society

2008 Gold Lecture, International Society of Presbyopia

2009 Fellow Award, Gold Level (FARVO), Association for Research in Vision and Ophthalmology

2010 Barraquer Award, International Society for Refractive Surgery of the American Academy of Ophthalmology

2011 Best Session Paper Award, 1st Joint International Congress, Refr@ctive.online and SICSSO, Grosseto, Italy

2013 IIRSI Gold Medal Award, Indian Intraocular Implant and Refractive Society, New Delhi, India

2014 Secretariat Award, American Academy of Ophthalmology

2016 Waring Medal for Editorial Excellence, *Journal of Refractive Surgery*

2018,'22,'24,'25 Gold Reviewer, *Journal of Refractive Surgery*

2019 – '21, '23 Silver Reviewer, *Journal of Refractive Surgery*

2023 The Visionary Award, American-European Congress of Ophthalmic Surgery. Deer Valley, Utah. July 15, 2023

2024 DeVaro S, Barros M, van den Berg RM, van den Berg AB, Bedi R, Klyce SD, Rocha KM. Estimating the effectiveness of small aperture IOL in reducing higher order

aberrations in post-radial keratotomy patients. Best of Paper Session – Residents and Fellows. Presenting Author: Sarah DeVaro, MD. 2024 ASCRS Annual Meeting, Boston, MA

2024 KCXL Recognition Award, International Keratoconus and CXL (KCXL) Experts' Meeting, December 12–14, 2024, Kongresshaus, Zurich, Switzerland.

2026 The Galileo Galilei Medal Lecture, the World Keratoconus Congress 2026, April 16-18, 2026, Florence, Italy.

PROFESSIONAL SOCIETY MEMBERSHIPS

Association for Research in Vision and Ophthalmology

International Society for Eye Research

New York Intra Ocular Lens Implant Society

American Society of Cataract and Refractive Surgery

Optica The Optical Society

International Intra-Ocular Implant Club

International Society of Refractive Surgery

Cornea Society

European Society of Cataract and Refractive Surgeons (ESCRS)

International Society for Contact Lens Research

Ophthalmic World Leaders

Pan-American Association of Ophthalmology

Yale Ophthalmological Society

BIBLIOGRAPHY

PUBLISHED ABSTRACTS

1. Stanley J, Mishima S, Klyce S: In vivo determination of endothelial permeability. Invest Ophthalmol 4:946 abs, 1965

2. Klyce SD, Dohlman CH: In vivo determination of corneal swelling pressure. Invest Ophthalmol 6:208 abs, 1967

3. Klyce SD, Tolpin DW, Goldman JN, Dohlman CH: Swelling properties of dogfish cornea. Invest Ophthalmol 7:229 abs, 1968

4. Klyce SD, Zadunaisky JA: Microelectrode profile of rabbit corneal epithelium. Biophys J 10:199 abs, 1970

5. Klyce SD, Zadunaisky JA: Location of the source of emf generated by the rabbit corneal epithelium. Biophys J 11:145 abs, 1971

6. Klyce SD: Microelectrode and tracer analysis of ion transport in the rabbit corneal epithelium. Abstracts of the IV International Biophysics Congress, Moscow. 3:106 abs, 1972

7. Klyce SD: Influence of Ag^+ on epithelial transport. Biophys J 16:131 abs, 1976

8. Wong RKS, Klyce SD: Cellular mechanisms underlying the epinephrine stimulation of Cl^- secretion in the rabbit corneal epithelium. Biophys J 17:24 abs, 1977

9. Burstein NL, Klyce SD: Electrophysiologic and morphologic effects of ophthalmic preparations on corneal epithelium. *Invest Ophthalmol Vis Sci* 16:1 abs, 1977

10. Klyce S, Beebe B, Bernegger O, Hu D: The influence of epithelial anoxia on corneal thickness in vitro. *Invest Ophthalmol Vis Sci* 16:2 abs, 1977

11. Klyce SD: Enhancing the stromal dehydrating properties of the corneal epithelial fluid pump. Symposium on Biological Transport Properties in the Eye. [invited lecture] *Invest Ophthalmol Vis Sci* 16:114 abs, 1977

12. Klyce SD, Wong RKS: Sympathetic influence on corneal epithelial electrophysiology. *Invest Ophthalmol Vis Sci* Suppl:190 abs, 1978

13. Klyce SD: Phenomenologic description of corneal hydration dynamics: A re-evaluation. *Invest Ophthalmol Vis Sci* Suppl:214 abs, 1978

14. Klyce SD, Bernegger O: Epithelial hypoxia, lactate production and stromal edema. *Invest Ophthalmol Vis Sci* Suppl:276 abs, 1978

15. Klyce SD, Russell SR: Cytoplasmic standing gradients could link epithelial Jv to solute transport. Biophys J 25:95 abs, 1979

16. Klyce SD: Cellular and stromal volume regulation in cornea. *Invest Ophthalmol Vis Sci* Suppl:104 abs, 1979

17. Klyce SD, Wong RKS: Adrenergic modulation of Cl^- transport across the corneal epithelium. Physiologist 22:70 abs, 1979

18. Klyce SD, Lambert SR: Sattler's veil: experimental demonstration and theory of origin. *Invest Ophthalmol Vis Sci* Suppl:62 abs, 1979

19. Klyce SD, Russell SR: The viscous flow theory: an adjunct to the standing gradient hypothesis. Federation Proc 39:378 abs, 1980

20. Klyce SD, Marshall WS: Membrane and paracellular pathway resistances in rabbit corneal epithelium. Intracellular activities and potentials in epithelia of the eye. [invited lecture] Proc Int Soc for Eye Research 1:89 abs, 1980

21. Klyce SD, Wong RKS, Beuerman RW: Sympathetic innervation of the rabbit cornea. Issues in the innervation of the cornea. [invited lecture] Proc Int Soc for Eye Research 1:86 abs, 1980

22. Marshall WS, Klyce SD: Membrane resistances in rabbit corneal epithelium. Federation Proc 40:370 abs, 1981

23. Klyce SD, Palkama KA, Marshall WS, Huhtaniitty S, Mann KP: Evidence for serotonergic receptors in corneal epithelium. *Invest Ophthalmol Vis Sci* Suppl 20:194 abs, 1981

24. Berkowitz RA, Klyce SD, Kaufman HE, Salisbury JD: Fluorophotometric determination of the corneal epithelial barrier after penetrating keratoplasty. *Invest Ophthalmol Vis Sci* Suppl 20:18 abs, 1981

25. Marshall WS, Klyce SD: Epithelial membrane resistance in the rabbit cornea. *Invest Ophthalmol Vis Sci* Suppl 20:195 abs, 1981

26. McDonald MB, Werblin TP, Kaufman HE, Klyce SD: Epikeratophakia - The surgical correction of aphakia: Predictability of correction. *Invest Ophthalmol Vis Sci* Suppl 20:68 abs, 1981

27. Marshall WS, Klyce SD: Control mechanisms of Cl^- secretion in some vertebrate epithelia. Abstracts of the IX International Symposium on Comparative Endocrinology, Hong Kong, 1981

28. Marshall WS, Klyce SD: Transport effects of serotonin in rabbit corneal epithelium. Federation Proc 41:1263 abs, 1982

29. Klyce SD, Russell SR: Water and nonelectrolyte transport in epithelia: modeling nonlinear systems. Federation Proc 41:1266 abs, 1982

30. Klyce SD: Symposium on cyclic nucleotides in the eye. Cyclic AMP control of transport in the cornea. [invited lecture] *Invest Ophthalmol Vis Sci* Suppl 22:13 abs, 1982

31. Marshall WS, Klyce SD: Serotonin stimulation of ion transport by the corneal epithelium: a microelectrode study. *Invest Ophthalmol Vis Sci* Suppl 22:100 abs, 1982

32. Safir A, McDonald MB, Klyce SD: Restoring donor corneas to normal dimensions and hydration before cryolathing. *Invest Ophthalmol Vis Sci* Suppl 22:241 abs, 1982

33. Klyce SD: Applications of nonlinear irreversible thermodynamics to hydration control in ocular tissues. Symposium on fluid transport and volume regulation in the eye. [symposium organizer] Proc V Int Cong Eye Res, Veldhoven, The Netherlands: 75 abs, 1982

34. Beuerman RW, Klyce SD, Kooner S, Tanelian D, Rozsa A: Dimensional analysis of rabbit ciliary nerve. *Invest Ophthalmol Vis Sci* Suppl 24:261 abs, 1983

35. Crosson CE, Rutter MHB, Klyce SD: Dopaminergic receptors in rabbit corneal epithelium. *Invest Ophthalmol Vis Sci* Suppl 24:67 abs, 1983

36. Klyce SD, Beuerman RW, Crosson CE, Rutter MHB: Superior cervical ganglionectomy effects on corneal transport. *Invest Ophthalmol Vis Sci* Suppl 24:67 abs, 1983

37. Crosson CE, Beuerman RW, Klyce SD: Dopamine modulation of corneal epithelial chloride transport via preterminal receptors. Federation Proc 43:1087 abs, 1984

38. McDonald MB, Yamaguchi T, Klyce SD, Kaufman HE: Epikeratophakia for the correction of myopia. *Invest Ophthalmol Vis Sci* 25:333 abs, 1984

39. Yamaguchi T, Asbell P, Ostrick M, Kimura T, Klyce S, McDonald M, Safir A, Kissling G, Kaufman HE: Long term followup of primate eyes after anterior radial keratotomy. *Invest Ophthalmol Vis Sci* Suppl 25:168 abs, 1984

40. Crosson CE, Beuerman RW, Klyce SD: Microelectrode study of membrane changes associated with sympathetic denervation in the rabbit corneal epithelium. *Invest Ophthalmol Vis Sci* Suppl 25:303 abs, 1984

41. McDonald MB, Klyce SD, Stephenson G, Suarez H, Kaufman HE: Myopic epikeratophakia. Ophthalmology Suppl 91:88 abs, 1984

42. Klyce SD: Computer-assisted analysis of human corneal topography. *Invest Ophthalmol Vis Sci* Suppl 25:167 abs, 1984

43. Klyce SD, Marshall WS, Crosson CE: Paracellular pathway in the rabbit corneal epithelium. Symposium on Paracellular Pathway and Junctions in Ocular Tissues. Proc Int Soc for Eye Research, 3:163 abs 1984

44. Crosson CE, Beuerman RW, Klyce SD: Electrophysiological correlates of sympathetic innervation in the rabbit corneal epithelium. Symposium on Sensory and Autonomic Innervation of the Eye. Proc Int Soc for Eye Research 3:70 abs, 1984

45. Klyce SD, Ochsner RD: Method for quantitative evaluation of corneal shape changes by contact lens molding. Symposium on the Effects of Contact Lenses on the Normal Physiology and Anatomy of the Eye. Proc Int Soc Eye Research 3:169 abs, 1984

46. Klyce SD, Crosson CE: Forskolin stimulation of rabbit corneal epithelial Cl^- transport. *Invest Ophthalmol Vis Sci* Suppl 26:1 abs, 1984

47. Crosson CE, Klyce SD, Cooper HD, McCollam GA, Beuerman RW: Corneal epithelial wound healing: A biphasic process. *Invest Ophthalmol Vis Sci* Suppl 26:177 abs, 1985

48. Justin N, Asbell PA, Klyce SD: The role of corneal shape in radial keratotomy. *Invest Ophthalmol Vis Sci* Suppl 26:149 abs, 1985

49. Beuerman RW, Klyce SD, Vigo MC, Dupuy BM, Crosson CE: Modulation of latency and velocity in corneal epithelial wound healing. *Invest Ophthalmol Vis Sci* Suppl 26:91 abs, 1985

50. Suarez E, McDonald M, Klyce S, Verity S, Salmeron B, Kaufman HE: Myopic epikeratophakia-new design. *Invest Ophthalmol Vis Sci* Suppl 26:203 abs, 1985

51. Klyce SD, Jenison GL, Crosson CE, Beuerman RW: Distribution of sympathetic nerves in the rabbit cornea. *Invest Ophthalmol Vis Sci* Suppl 27:354 abs, 1986

52. Pedroza L, Beuerman RW, Klyce SD, Crosson CE: Keratocyte degeneration following epithelial abrasion. *Invest Ophthalmol Vis Sci* Suppl 27:31 abs, 1986

53. Crosson CE, Klyce SD, Bazan HEP, Bazan NG: Protein kinase C activation in the modulation of Cl^- transport by the corneal epithelium. *Invest Ophthalmol Vis Sci* Suppl 27:82 abs, 1986

54. Maguire LJ, Singer DE, Klyce SD: Improved high resolution graphic presentation of computer analyzed keratoscope photographs. *Invest Ophthalmol Vis Sci* Suppl 27:15 abs, 1986

55. Beuerman RW, Crosson CE, Klyce SD: Formation of the leading edge in corneal reepithelialization. Scanning Electron Microscopy Meeting abs, 1986

56. Maguire LJ, Klyce SD, McDonald MB, Kaufman HE: Computer analysis of keratoscope photographs; corneal topography after radial keratotomy. Am Acad Ophthalmol, Ophthalmology Suppl 93:115, 1986

57. Dingeldein SA, Klyce SD: Computer assisted corneal topography of normal corneas. *Invest Ophthalmol Vis Sci* Suppl 28:222, 1987

58. Klyce SD: Cl^- transport in rabbit corneal epithelium stimulated by vasoactive intestinal polypeptide. *Invest Ophthalmol Vis Sci* Suppl 28:375, 1987

59. Reinach P, Bazan HEP, Klyce SD, Bazan NG: Effects of phorbol esters on active ion transport in frog corneal epithelium. *Invest Ophthalmol Vis Sci* Suppl 28:375, 1987

60. Klyce SD, Bonanno JA: Role of the epithelium in corneal hydration. Third World Congress on the Cornea. Cornea 6:152, 1987

61. Klyce SD, Dingeldein SA: Imaging of the cornea. Third World Congress on the Cornea. Cornea 6:152, 1987

62. Dingeldein SA, Klyce SD, McDonald MB, Kaufman HE: Corneal topographic stability following myopic epikeratophakia. Am Acad Ophthalmol, Ophthalmology Suppl 94:125, 1987

63. Klyce SD, Dingeldein SA, Bonanno JA, McDonald MB, Kaufman HE: Hydrogel implants: Evaluation of first human trial. *Invest Ophthalmol Vis Sci* Suppl 29:393, 1988

64. Kwok LS, Klyce SD: Age-related changes in corneal stromal hydration: Quantitative performance of the Donnan swelling theory. *Invest Ophthalmol Vis Sci* Suppl 29:174, 1988

65. McDonald M, Frantz J, Santana E, Salmeron B, Stephenson G, Klyce S, Beuerman R, Kaufman H: Excimer laser surface shaping of the primate cornea for the correction of myopia. *Invest Ophthalmol Vis Sci* Suppl 29:310, 1988

66. Bonanno JA, Klyce SD: Effects of furosemide and low [Na] or [Cl] on stimulated Cl^- transport in rabbit corneal epithelium. *Invest Ophthalmol Vis Sci* Suppl 29:174, 1988

67. Dingeldein SA, Pittman SD, Wang J, Klyce SD: Analysis of corneal topographic data. *Invest Ophthalmol Vis Sci* Suppl 29:389, 1988

68. Klyce SD, Dingeldein SA, Pittman SD, Wang J: Computer-assisted analysis of topographic data in human corneas. Am Acad Ophthalmol, Ophthalmol Suppl 95:163, 1988.

69. Kaufman HE, McDonald MB, Insler MS, Klyce SD: Cornea and refractive surgery. Am Acad Ophthalmol, Ophthalmology Suppl 95:231, 1988.

70. Wang J, Rice DA, Klyce SD: Investigation and improvement of corneal topographical analysis. IEEE Engineering in Medicine and Biology Society, 10th International Conference, 1988.

71. Crosson CE, Klyce SD: Paracellular pathway of the rabbit corneal epithelium. 8th International Congress of Eye Research 5:149, 1988.

72. Beuerman RW, McDonald MB, Frantz JM, Santana E, Salmeron B, Stephenson G, Klyce SD, Kaufman HE: Wound healing following excimer laser ablation in primates. 8th International Congress of Eye Research 5:2, 1988.

73. Lehtosalo J, Uusitalo RJ, Klyce SD: Total corneal flattening produced by epikeratophakia for keratoconus. 8th International Congress of Eye Research 5:113, 1988.

74. Klyce SD: Endothelial permeability and fluid transport revisited. 8th International Congress of Eye Research 5:93, 1988.

75. Klyce SD, Pittman SD, Wang J: Problems in measuring and imaging the cornea. 8th International Congress of Eye Research 5:61, 1988.

76. Klyce SD, Wilson SE, Pittman SD, Wang J: Estimation of potential visual acuity from corneal shape. *Invest Ophthalmol Vis Sci* Suppl 30:218, 1989.

77. Lin DTC, Wilson SE, Klyce SD, Insler MS: Post keratoplasty corneal topographic changes following suture removal quantified by the Corneal Modeling System. *Invest Ophthalmol Vis Sci* Suppl 30:188, 1989.

78. Wilson SE, Klyce SD, Insler MS, Lin DTC: Contact lens-induced changes in corneal topography. *Invest Ophthalmol Vis Sci* Suppl 30:218, 1989.

79. Kwok LS, Klyce SD: The origin of the swelling pressure in the rabbit corneal stroma. *Invest Ophthalmol Vis Sci* Suppl 30:168, 1989.

80. McDonald M, Shofner S, Klyce S, Frantz J, Munnerlyn C, Clapham T, Kaufman H: Clinical results of central photorefractive keratectomy (PRK) with the 193 nm excimer laser for the treatment of myopia: The blind eye study. *Invest Ophthalmol Vis Sci* Suppl 30:216, 1989.

81. Klyce SD: Imaging, reconstruction, and display of corneal topography. SPIE 33rd Annual International Technical Symposium of Optical and Optoelectronic Applied Science and Engineering, in press.

82. Klyce SD: Interpretation and clinical uses of corneal topographical analysis. Ophthalmology '89, Annual Meeting of the Saudi Ophthalmological Society, Jeddah, Saudi Arabia, March 29-April 1, 1989.

83. Kwok LS, Klyce SD: Effect of hydrogen peroxide on epithelial ion transport in rabbit cornea. Abstracts of the Cornea Research Conference, Boston, MA, September 23, 1989

84. Uusitalo RJ, Lehtosalo J, Klyce SD: Two year follow-up of epikeratophakia for keratoconus. XXIX:e Nordiska Ögonläkarmötet, Reykjavik, Iceland, June 18-21, 1989.

85. Klyce SD: Current technology available for evaluating corneal topography - how predictive and how simple to use? Abstracts of the Seventh Congress of the European Intraocular Lens Implant Council, Zürich, Switzerland, August 27-31, 1989.

86. Wilson SE, Lin DTC, Klyce SD, Reidy JJ, Insler MS: Topographic changes in contact lens-induced corneal warpage, Ophthalmology 96:122, 1989.

87. Lin DTC, Wilson SE, Reidy JJ, Klyce SD, McDonald MB, Kaufman HE: Minimal postkeratoplasty astigmatism using an adjustable suture. Ophthalmology 96:120, 1989.

88. McDonald MB, Shofner RS, Klyce SD, Frantz JM, Munnerlyn C, Clapham T, Kaufman HE: Excimer laser photorefractive keratectomy (PRK) for myopia: the blind eye study. Ophthalmology 96:98, 1989.

89. Kwok LS, Klyce SD: Effect of hydrogen peroxide on the corneal epithelial barrier. American Academy of Optometry Annual Meeting Program, 111, 1989.

90. Klyce SD: Corneal topography. Abstracts of the XXVI International Congress of Ophthalmology, 122, 1990.

91. Klyce SD: Advanced Imaging Technology in Ophthalmology. Abstracts of the XXVI International Congress of Ophthalmology, 129, 1990.

92. Kwok LS, Klyce SD: Effect of t-butyl hydrogen peroxide on rabbit corneal epithelium. *Invest Ophthalmol Vis Sci* Suppl 31:360, 1990.

93. Wilson SE, Klyce SD, McDonald MB: Correlation of manifest refraction and corneal topography following excimer laser photorefractive keratectomy. *Invest Ophthalmol Vis Sci* Suppl, 31:245, 1990.

94. Klyce SD, Wang J, Rice DA: Errors associated with the reconstruction of the corneal surface from keratoscopy. *Invest Ophthalmol Vis Sci* Suppl 31:480, 1990.

95. AbdelMageed MA, McDonald MB, Klyce SD, Steinert R, Waring GO, Knight P: Postoperative Corneal Topography of Aphakic Human Eyes with Hydrogel Intracorneal Implants. *Invest Ophthalmol Vis Sci* Suppl 31:301, 1990.

96. Salmeron B, McDonald MB, Beuerman RW, Klyce SD, Pedroza L, Kaufman HE: Long-term (5-Year) biocompatibility and histology of a hydrogel intrastromal implant in a rhesus monkey eye. *Invest Ophthalmol Vis Sci* Suppl 31:29, 1990.

97. McDonald MB, Liu J, Andrade H, AbdelMageed M, Varnell R, Salmeron B, Klyce SD, Sunderland G, Munnerlyn C, Clapham T, Kaufman HE: Clinical results of 193-nm excimer laser central photorefractive keratectomy (PRK) for myopia: The partially sighted and sighted eye studies. *Invest Ophthalmol Vis Sci* Suppl 31:245, 1990.

98. Klyce SD: Cell volume regulation in the cornea. Proc Int Soc for Eye Research 6:28 abs, 1990.

99. Klyce SD, Wilson SE, McDonald MB: Corneal topography after photorefractive keratectomy. Proc Int Soc for Eye Research 6:134 abs, 1990.

100. McDonald M, Liu J, AbdelMegeed, Andrade H, Varnell R, Salmeron B, Klyce S, Munnerlyn C, Clapham T, Kaufman H: 193-nm excimer laser central photorefractive keratectomy (PRK) for myopia: partially sighted and sighted eyes. Proc Int Soc for Eye Research 6:134 abs, 1990.

101. Klyce SD: Corneal Topography. Proc Int Soc for Eye Research 6:211 abs, 1990.

102. Uusitalo RJ, Lehtosalo J, Klyce SD: Topographical analysis of corneal flattening produced by epikeratophakia for keratoconus. Proc Int Soc for Eye Research 6:192 abs, 1990.

103. Klyce SD: New methodologies of investigation: Corneal topography. ISCLR Sixth Scientific Meeting, Monte Carlo, August 30 - September 3, 1990.

104. Kwok LS, Klyce SD: Tear-side t-butyl hydrogen peroxide alters epithelial thickness in the rabbit cornea. American Academy of Optometry Annual Meeting Program, 1990.

105. Wilson SE, Klyce SD, McDonald MB, Liu JC, Kaufman HE: Corneal topographic alterations produced by excimer laser myopic photorefractive keratoplasty. Ophthalmology (Suppl) 97:127, 1990

106. McDonald MB, Liu JC, AbdelMegeed M, Andrade HA, Klyce SD, Varnell R, Munnerlyn C, Clapham T, Salmeron B, Kaufman HE: Results of excimer laser photorefractive keratectomy and phototherapeutic keratectomy on 42 patients. Ophthalmology (Suppl) 97:127, 1990.

107. Friedman RS, Wilson SE, Klyce SD: Contact lenses to mold corneal topography after penetrating keratoplasty. *Invest Ophthalmol Vis Sci* Suppl 32:721, 1991.

108. Klyce SD, Wilson SE, McDonald MB, Liu JC, Kaufman HE: Corneal topography after excimer laser keratectomy. *Invest Ophthalmol Vis Sci* Suppl 32:778, 1991.

109. Klyce SD: Application of corneal topography in refractive surgery. Abstracts of the Fourth International Cataract, Implant, Microsurgical and Refractive Keratoplasty Meeting, 70, 1991.

110. Klyce SD: Corneal topographical analysis. Clinical applications. Abstracts of the 24th Panhellenic Ophthalmological Congress, 1991.

111. Klyce SD: The topography of phototherapeutic keratectomies and photorefractive keratectomies using the 193-nm excimer laser. Abstracts of the 6th International Congress on Cataract and Refractive Surgery. Florence, Italy, July 4-7, 1991.

112. Klyce SD: Modern corneal topography analysis: corneal astigmatism. Abstracts of the 6th International Congress on Cataract and Refractive Surgery. Florence, Italy, July 4-7, 1991.

113. McDonald M, Leach D, Salinas R, Varnell R, Byrd T, Lehman E, Klyce S, Kaufman H: Excimer update: PRK, PTK, experimental hyperopia data. Abstracts of the Annual International Society for Refractive Keratoplasty Meeting, Anaheim, CA, 1991.

114. Klyce, SD, Smolek MK, Wilson SE, McDonald M: Topography of PRK for myopia: update. Abstracts of the Annual International Society for Refractive Keratoplasty Meeting, Anaheim, CA, 1991.

115. Klyce, SD: Application of modern corneal topography analysis. Abstracts of the European Research Symposium, Geneva, Switzerland, 1991.

116. Klyce SD: Corneal topography principles. Abstracts of the Excimer Laser Meeting, Heraklion, Greece, 1991.

117. Klyce SD: Corneal topography in excimer laser refractive surgery. Abstracts of the Excimer Laser Meeting, Heraklion, Greece, 1991.

118. Ruiz-Montenegro JA, Mafra CH, Jumper JM, Wilson SE, Klyce SD: Corneal topography in asymptomatic, normal wearers of soft and rigid contact lenses. Abstracts of the Castroviejo Cornea Society, 1991.

119. Mafra CH, Ruiz-Montenegro JA, Jumper JM, Wilson SE, Klyce SD: Corneal topography in asymptomatic normal wearers of soft and rigid contact lenses. *Invest Ophthalmol Vis Sci* Suppl 33:697, 1992

120. Klyce SD, Estopinal HA, Gersten M, Gormley DJ, Moore JW: Fluorescein exam simulation for contact lens fitting. *Invest Ophthalmol Vis Sci* Suppl 33:697, 1992.

121. Smolek MK, Klyce SD: Corneal stromal cohesion strength: demonstration of structural anisotropy. *Invest Ophthalmol Vis Sci* Suppl 33:895, 1992.

122. Klyce SD, Smolek MK, Wilson SE, McDonald MB: Topography of the excimer laser ablation. Proc Int Soc for Eye Research VII, Exptl Eye Res (Suppl) 55:88, 1992.

123. Klyce SD: Videokeratography: Application of 3-D reconstructions in the evaluation of corneal optical performance. Proc Int Soc for Eye Research VII, Exptl Eye Res (Suppl) 55:135, 1992.

124. Klyce SD: Application of corneal topography to corneal transplantation. Proc Xth Afro-Asian Cong Ophthalmol, Jakarta, Indonesia, July 17-21, 1992, p 34.

125. Klyce SD: Clinical applications of corneal topography in refractive surgery. Proc Xth Afro-Asian Cong Ophthalmol, Jakarta, Indonesia, July 17-21, 1992, p 57.

126. Klyce SD: Corneal topography in health and disease. Proc Xth Afro-Asian Cong Ophthalmol, Jakarta, Indonesia, July 17-21, 1992, p 67.

127. Klyce S, Smolek MK, Maeda N, McDonald MB: Topographic changes after excimer PRK. Abstracts of the 1992 Pre-Academy Meeting of the ISRK, November 7, 1992, Dallas.

128. Maeda N, Klyce SD, Dehkhoda MT, Smolek MK: Automated keratoconus detection with corneal topography analysis. 1992 Annual Scientific Meeting of the Castroviejo Cornea Society, Dallas, TX. Cornea 12:88, 1993.

129. Maeda N, Klyce SD, Hamano H: Alteration of corneal asphericity in RGP lens-induced warpage. 1993 CLAO Annual Meeting Program, p 60.

130. Klyce SD, Rabinowitz YS, Wilson SE: Understanding corneal topography. 1993 CLAO Annual Meeting Program, p 46.

131. Klyce SD, Updegraff SA, Smolek MK, McDonald MB, Zadnik K, Mutti DO, Trokel SL: Predictability in keratorefractive surgery. *Invest Ophthalmol Vis Sci* 34:801, 1993.

132. Hershel MK, McDonald MB, Ahmed S, Klyce SD, Varnell RJ, Thompson HW: Voltaren for treatment of discomfort after excimer ablation. *Invest Ophthalmol Vis Sci* 34:893, 1993.

133. McDonald MB, Hershel MK, Ahmed S, Varnell RJ, Klyce SD, Beuerman RW: Effects of voltaren and FML drops in post-excimer laser photorefractive keratectomy therapy. *Invest Ophthalmol Vis Sci* 34:803, 1993.

134. Russell RS, McDonald MB, Klyce SD, Varnell RJ: Excimer laser ablation rate in corneal stroma. *Invest Ophthalmol Vis Sci* 34:801, 1993.

135. Maeda N, Klyce SD, Smolek MK, Thompson HW: Automated keratoconus screening with corneal topography analysis. *Invest Ophthalmol Vis Sci* 34:1218, 1993.

136. Smolek MK, Klyce SD, Maeda N, McDonald MB: Serial event analysis using statistical reduction of corneal topography. *Invest Ophthalmol Vis Sci* 34:1251, 1993.

137. Parker PJ, Klyce SD, Ryan BL, Lindstrom RL, Sher NA, Lane SS, Carpel EC, Ostrov C, DJ Doughman: Central topographic islands following photorefractive keratectomy. *Invest Ophthalmol Vis Sci* 34:803, 1993.

138. Mafra CH, Dave A, Wilson SE, Klyce SD: Extracapsular cataract surgery: corneal topographic alterations. *Invest Ophthalmol Vis Sci* 34:1248, 1993.

139. Husseini Z, Wilson SE, Klyce SD: Standardized scales for corneal topographic maps. *Invest Ophthalmol Vis Sci* 34:1252, 1993.

140. Gardner BP, Klyce SD, Thompson HW, Lin DTC, Hodkin MJ, McDonald MB: Centration of photorefractive keratectomy: topographic assessment. *Invest Ophthalmol Vis Sci* 34:803 1993.

141. Klyce SD: Videokeratoscopy in contact lens practice. The Holland Guest Lecture. Abstracts of the 23rd Congress of the European Contact Lens Society of Ophthalmologists, Amsterdam, Nederlands, May 20-23, 1993, p 60.

142. Klyce SD: Update on excimer topography issues. Abstracts of the ISRK 1993 Mid-summer Symposium, Minneapolis, Minnesota, July 23-25, 1993, p26.

143. Klyce SD: Corneal topography: why the refractive cataract surgeon won't be able to live without it. Abstracts of the Second Annual Ocular Surgery News Symposium, Small Incision & Refractive Cataract Surgery, New York, New York, Sept 11-12, 1993, p17.

144. Klyce SD, Maeda N, Smolek MK: Use of neural networks in interpretation of videokeratography. Abstracts of the ISRK Pre-Academy Refractive Surgery Workshop, Chicago, Illinois, November 12-13, 1993.

145. Klyce SD, Maeda N, Smolek MK: Use of neural networks for interpretation of corneal topography: I. Initial demonstration. *Invest Ophthalmol Vis Sci* 35:2079, 1994.

146. Smolek MK, Klyce SD, Maeda N: Use of neural networks for interpretation of corneal topography: II. A data simulation paradigm for optimization. *Invest Ophthalmol Vis Sci* 35:2079, 1994.

147. Maeda N, Klyce SD, Smolek MK, Rabinowitz YS: Comparison of three keratoconus detection schemes with videokeratography. *Invest Ophthalmol Vis Sci* 35:2078, 1994.

148. Pillai CT, Bowman RW, Wilson SE, Klyce SD: Changes in corneal topography after removal of rigid contact lenses in candidates for refractive surgery. *Invest Ophthalmol Vis Sci* 35:1739, 1994.

149. McDonald MB, Telfair WB, Nesburn AB, Salz JJ, Macy JI, Pender PM, Kearney WF, Pendleton KM, Klyce SD, Varnell RJ: Excimer laser hyperopia PRK phase I: The blind eye study. *Invest Ophthalmol Vis Sci* 35:1488, 1994

150. Moreno ML, McDonald MB, Ribeiro J, Klyce SD, Smolek M: Effect of decentration of the ablation on contrast sensitivity after excimer photorefractive keratectomy for mild to moderate myopia. *Invest Ophthalmol Vis Sci* 35:2141, 1994.

151. Maeda N, Klyce SD, Smolek MK, McDonald MB: Disparity of corneal power within the pupil and keratometry readings after refractive surgery for myopia. Ophthalmology (Suppl) 101:104, 1994.

152. Moreno LC, McDonald MB, Ribeiro JC, Klyce SD, Smolek M: Contrast sensitivity of decentered photorefractive keratectomies for myopia. Abstracts of the 1994 Pre-Academy Meeting of the ISRK, San Francisco, California, October 28-29, 1994, p 54.

153. Ribeiro JC, McDonald MB, Klyce SD: Excimer laser photorefractive keratectomy (PRK) resulting in more myopia postoperatively than preoperatively. Abstracts of the 1994 Pre-Academy Meeting of the ISRK, San Francisco, California, October 28-29, 1994.

154. Maeda N, Klyce SD, Smolek MK, McDonald MB: Topographic correlates to visual acuity after photorefractive keratectomy (PRK). Abstracts of the 1994 Pre-Academy Meeting of the ISRK, San Francisco, California, October 28-29, 1994, p 59.

155. Macy JI, Telfair W, Mahdavi S, Shimmick J, Nesburn A, Salz J, McDonald M, Klyce S, Pendleton K, Varnell R, Pender P, Kearney W: Photorefractive keratectomy (PRK) for hyperopia: Phase I. Abstracts of the 1994 Pre-Academy Meeting of the ISRK, San Francisco, California, October 28-29, 1994, p 109.

156. Lemos MM, Nevitt MP, McDonald MB, Ribeiro JC, Swinger CA, Pinheiro MN, McDonnell PJ, Itoi M, Aquavella JV, Klyce SD, Varnell RJ, LeBlanc B: Decentration of the ablation zone in photorefractive keratectomy comparing two fixation techniques: a multicenter study. Abstracts of the 1994 Pre-Academy Meeting of the ISRK, San Francisco, California, October 28-29, 1994, p 59.

157. Klyce SD, Maeda N, Smolek MK: Corneal topography. Abstracts of the XXIV Congress of the European Contact Lens Society of Ophthalmologists, Alicante, Spain, Sept 14-17, 1994, p73.

158. Klyce SD, Maeda N, Smolek MK: Corneal topography analysis: the expert systems approach. Abstracts of the Aegean Cornea II Meeting, Rhodes, Greece, Sept 30-Oct 2, 1994, p43.

159. Alimisi S, Miltsakakis D, Klyce S: Corneal topography use in patients with irregular or incomplete keratometer mires undergoing cataract extraction. Case Reports. Abstracts of the Aegean Cornea II Meeting, Rhodes, Greece, Sept 30-Oct 2, 1994, p44.

160. Bucci FA, Moody KJ, Tanner JB, Klyce SD: Corneal topographic analysis of ring-shaped irregular astigmatism caused by annular tinted contact lenses. 1995 CLAO Annual Meeting Program, p 47.

161. Klyce SD, Maeda N, Smolek MK, McDonald MB: Evaluating refractive surgery with corneal topography. *Invest Ophthalmol Vis Sci* 36: S1, 1995

162. Holland D, Maeda N, Hannush SB, Riveroll L, Klyce SD, Wilson SE: Unilateral keratoconus: incidence and comparison of quantitative topographic indices with bilateral keratoconus. *Invest Ophthalmol Vis Sci* 36: S308, 1995.

163. Hargrave S, Kalin NS, Maeda N, Klyce SD, Wilson SE: Automated topographic screening for keratoconus in refractive surgery candidates. *Invest Ophthalmol Vis Sci* 36: S309, 1995.

164. Smolek MK, Klyce SD, Maeda N: Time lapse videokeratography using computer morphing. *Invest Ophthalmol Vis Sci* 36: S380, 1995.

165. Nevitt M, McDonald M, Chitkara D, Beuerman R, Klyce S, Frey R, Downes D: Corneal wound healing after photorefractive keratectomy with the autonomous technologies scanning & tracking excimer laser: a histopathologic study in a non human primate model. *Invest Ophthalmol Vis Sci* 36: S710, 1995.

166. McDonald M, Nevitt M, Chitkara D, Klyce S, Pendleton K, Liang K, Frey R, Downes R: First clinical evaluation of autonomous technologies scanning and tracking excimer laser a primate study. *Invest Ophthalmol Vis Sci* 36: S710, 1995.

167. Klyce SD, Maeda N, Smolek MK: Corneal indices and neural networks for the interpretation of corneal topography. European J Ophthalmol 5 (Suppl):163, 1995.

168. Klyce SD: Videokeratoscopy in contact lens practice. Keynote lecture. Abstracts of the XXIV Congress of the European Contact Lens Society of Ophthalmologists, Clermont Ferrand, France, Sept 29-Oct 1, 1995.

169. Klyce SD: Comparison of irregular astigmatism: RK, PRK, ALK, and LASIK. Abstracts of the IV Annual Ocular Surgery News Symposium, New York City, Sept 16-17, 1995, p 151.

170. Bucci FA, Jr, Klyce SD: Management of eightball hyphema, iatrogenic aniridia, and traumatic cataract following severe blunt trauma in a patient with previous mini-RK/AK. Abstracts of the 1995 Pre-Academy Meeting of the ISRS, Atlanta, GA, October 26-28, 1995, p 78-79.

171. Lucci L, Martinez C, Klyce SD, McDonald MB, Lynn MJ, Waring GO, III, and the PERK Study Group: Topographic correlates to diurnal variations in vision in a 10 year PERK cohort. Abstracts of the 1995 Pre-Academy Meeting of the ISRS, Atlanta, GA, October 26-28, 1995, pp 116-117.

172. Maeda N, Klyce SD, Smolek MK: Comparison of methods for detecting keratoconus using videokeratography. Ophthalmol Digest 1:3-4, 1996.

173. Klyce SD, Lynn MJ, Smolek MK, McDonald MB, Waring GO III, PERK Study Group: Topography of PERK corneas at 10 years. *Invest Ophthalmol Vis Sci* 37:S933, 1996.

174. Smolek MK, Klyce SD: Neural network pair for the detection and grading of keratoconus suspects by videokeratoscopy. *Invest Ophthalmol Vis Sci* 37:S910, 1996.

175. Ruberti JW, Klyce SD: New computational model of corneal hydration dynamics. *Invest Ophthalmol Vis Sci* 37:S352, 1996.

176. Martinez CE, Applegate RA, Howland HC, Klyce SD, McDonald MB, Medina JP: Changes in corneal aberration structure after photorefractive keratectomy. *Invest Ophthalmol Vis Sci* 37:S933, 1996.

177. Maeda N, Klyce SD, Watanabe H, Kiritoshi A, Shimomura Y, Hamano H, Tano Y: Artificial neural networks in videokeratography for the screening of refractive surgery candidates. *Invest Ophthalmol Vis Sci* 37:S911, 1996.

178. Maeda N, Klyce SD: Practical use of automated keratoconus detection program in the screening of refractive surgical candidates. Abstracts of the 1996 Mid-Summer Symposium of the ISRS, Minneapolis, Minnesota, July 19-21, 1996, p 60.

179. Klyce SD: Corneal optical performance after refractive surgery. Abstracts of the Aegean Cornea III Meeting, Samos Island, Greece, September 6-8, 1996, p 31.

180. Martinez CE, Applegate RA, Howland HC, Klyce SD, McDonald MB, Medina JP: Photorefractive keratectomy and the monochromatic aberrations of the cornea. Abstracts of the 1996 Pre-Academy Meeting of the ISRS, Chicago, IL, October 26-28, 1996.

181. Klyce SD: Topographic comparison of refractive surgical procedures. Japan J Ophthalmic Surg 10:154, 1997

182. Ruberti JW, Klyce SD: Evidence of corneal endothelial membrane permeability and transport modulation by osmotic shock. *Invest Ophthalmol Vis Sci* 38:S683, 1997

183. Klyce SD, McDonald MB, Doubrava M, Kim C, Tan A, Srivannaboon S: Small beam tracking excimer laser PRK topography. *Invest Ophthalmol Vis Sci* 38:S697, 1997

184. Smolek MK, Klyce SD, Brenner DS: Zernike polynomial terms and corneal indexes as neural network inputs for videokeratography classification. *Invest Ophthalmol Vis Sci* 38:S920, 1997

185. Srivannaboon S, McDonald MB, Doubrava M, Klyce SD: A prospective clinical trial comparing a topographically guided artificial intelligence software system versus clinical expertise for fitting normal and pathological corneas with contact lenses. *Invest Ophthalmol Vis Sci* 38:S1089, 1997

186. Ruberti JW, Klyce SD: A computational transport model for membrane bound soft tissue. FASEB J 11(3):A20, 1997

187. Klyce SD: The role of videokeratoscopy in contact lens practice. Abstracts of the 40th annual meeting of the Japan Contact Lens Society, June 13-15, 1997. Osaka, Japan.

188. Klyce SD: Understanding the topography of laser refractive surgery. Abstracts of Advanced Ophthalmic Laser Surgery, June 26-28, 1997. Interlaken, Switzerland.

189. Klyce SD: Preoperative evaluation of refractive surgery. Abstracts of Advanced Ophthalmic Laser Surgery, June 26-28, 1997. Interlaken, Switzerland.

190. Klyce SD: [invited lecture] Role of corneal topography in refractive surgery. Abstracts of the 27th Congress of the European Contact Lens Society of Ophthalmologists, July 30 - August 2, 1997. Tampere, Finland.

191. McDonald MB, Salz JJ, Deitz MR, Frantz JM, Krueger RR, Kraff MC, Klyce SD, Scheinblum KA, Mutyala S, Stone JR: Phase III results on spherical myopia and myopia with astigmatism with the Autonomous Technologies (ATC) Ladarvision excimer laser system. *Invest Ophthalmol Vis Sci* 39:S350, 1998.

192. Smolek MK, Klyce SD: Surface area of the cornea appears to be conserved in keratoconus. *Invest Ophthalmol Vis Sci* 39:S636, 1998.

193. Ruberti JW, Klyce SD: Investigation of corneal endothelial response to NaCl osmotic perturbations. *Invest Ophthalmol Vis Sci* 39:S906, 1998.

194. Klyce SD , Smolek MK , Oshika T , Coorpender SJ , Maeda N , Haight DH , McDonald MB, Doubrava MW , Kim CK, Tan AL , Srivannaboon S: Corneal topography after PRK**.** Abstracts of the XXVIIIth Int Cong Ophthalmology, June 21-26, 1998, Amsterdam, Holland, 92-93.

195. Klyce SD: Comparison of topographical machines as an aid in contact lens fitting**.** Abstracts of the International Medical Contact Lens Symposium, June 23-25, 1998, Amsterdam, Holland, 44.

196. Klyce SD, Ruberti JW: Epithelial debridement alters stromal properties. Abstracts of Aegean Cornea IV, June 26-28, 1998, Santorini Island, Greece, 66.

197. Ruberti JW, Klyce SD: Dynamics of the corneal stroma. [Invited lecture] XIII I.C.E.R., July 26-31, Paris, France. Exptl Eye Res 67(S1):S206, 1998.

198. Klyce SD: Modern advances in detection and diagnosis of keratoconus using corneal topography. [Invited lecture] XIII I.C.E.R., July 26-31, Paris, France. Exptl Eye Res 67(S1):S209, 1998.

199. Klyce SD: Development of modern corneal topography. Abstracts of the 30th Annual Scientific Congress for the Royal Australian College of Ophthalmologists, Oct 11-15, 1998, Gold Coast, Queensland, Australia, p 37.

200. Klyce SD: Topographic evaluation of optical quality after refractive surgery. Abstracts of the 30th Annual Scientific Congress for the Royal Australian College of Ophthalmologists, Oct 11-15, 1998, Gold Coast, Queensland, Australia, p 125.

201. Klyce SD: Pearls for the successful use of corneal topography in your practice. Abstracts of the 30th Annual Scientific Congress for the Royal Australian College of Ophthalmologists, Oct 11-15, 1998, Gold Coast, Queensland, Australia, p 146.

202. Endl MJ, Martinez CE, Klyce SD, McDonald MB, Coorpender SJ, Applegate RA, Howland HC: Corneal aberrations after photorefractive keratectomy using a 5.5 mm ablation zone with a 7 mm transition zone. *Invest Ophthalmol Vis Sci* 40:S7, 1999.

203. Klyce SD, Karon MD, Ruberti JW, Smolek MK, Goto T: Epithelial debridement initiates acute anterior stromal edema. *Invest Ophthalmol Vis Sci* 40:S193, 1999.

204. Smolek MK, Klyce SD: Demonstration of a wavelet-based neural network for recognition of corneal topography patterns after myopic refractive surgery. *Invest Ophthalmol Vis Sci* 40:S176, 1999.

205. Ruberti JW, Gong H, Freddo TF, Klyce SD, Johnson M: Outward flow enhances large mobile molecule concentration gradients in corneal stroma. *Invest Ophthalmol Vis Sci* 40:S82, 1999.

206. Smolek MK, Klyce SD: Demonstration of a potentially universal, cross-platform corneal topography analysis method for screening. *Invest Ophthalmol Vis Sci* 41:S677 2000.

207. Klyce SD: Corneal asphericity and contact lens wear. *Abstracts of the 24th Annual Clinical Conference and Exhibition.* May 26-28, 2000. Birmingham, England, p 21.

208. Klyce SD: Corneal topography in wavefront-guided keratorefractive surgery. *Abstracts of Aegean Cornea V.* June 30 - July 2, 2000. Heraklion, Crete, Greece, p 13.

209. Klyce SD: Measuring the anterior surface of the cornea using corneal topography. *Abstracts of the 98th Tagung der Deutschen Ophthalmologischen Gesellschaft.* Berlin 21-24 September 2000.

210. Klyce SD: Corneal topography in the 21st century. *Abstracts of the 30th Congress of the European Contact Lens Society of Ophthalmologists*. Antalya, Turkey, September 27-30. *Contactologia* 22:119, 2000.

211. Klyce SD: Corneal topography applications in refractive surgery. *Abstracts of the 30th Congress of the European Contact Lens Society of Ophthalmologists*. Antalya, Turkey, September 27-30. *Contactologia* 22:119-120, 2000.

212. Klyce SD: Corneal topography and beyond. *Proceedings of the International Society for Eye Research, Volume XIV.* October 15-20, *Exptl Eye Res* 71:S70, 2000.

213. Smolek MK, Klyce SD: Corneal topography prototypes: maps, power spectra, and caustic dimensions. *Invest Ophthalmol Vis Sci*, 42:S528, 2001.

214. Karon MD, Klyce SD, Schilleci JT, Ruberti JW: Effect of inhibition of inflammatory mediators on trauma induced stromal edema. *Invest Ophthalmol Vis Sci* 42:S902, 2001

215. Karon MD, Klyce SD, Schilleci JT, Ruberti JW: Corneal stromal swelling dynamics after epithelial debridement. BED-Vol 50, 2001 Bioengineering Conference. ASME 2001

216. Klyce SD: Fundamentals of topography and aberrometry. *J Refractive Surg* 17:S265, 2001

217. Smolek MK, Klyce SD: Reliability of zernike polynomials in corneal and wavefront analysis. ARVO Abstract #3943, Association for Research in Vision and Ophthalmology, Fort Lauderdale, Florida, May 5-10, 2002. [for text, see http://www.arvo.org]

218. Cerviño A, McDonald, MB, Klyce SD: Shack-Hartmann pattern as an aid in the diagnosis and evaluation of dry eye: a retrospective study. ARVO Abstract #2029, Association for Research in

Vision and Ophthalmology, Fort Lauderdale, Florida, May 5-10, 2002. [for text, see http://www.arvo.org]

219. Smolek MK, Klyce SD: Standard absolute scales for corneal wavefront error maps. ARVO Abstract #2565, Association for Research in Vision and Ophthalmology, Fort Lauderdale, Florida, May 2-5, 2003. [for text, see http://www.arvo.org]

220. Klyce S: A tribute to David Maurice. ARVO Abstract #15, Association for Research in Vision and Ophthalmology, Fort Lauderdale, Florida, May 2-5, 2003. [for text, see http://www.arvo.org]

221. Klyce SD: Corneal topography and wavefront: clinical utility. [Invited lecture] *Abstracts of the 18th Meeting of the Japanese Society of Cataract and Refractive Surgery*, p 77, Kyoto, June 27-29, 2003.

222. Smolek MK, Klyce SD, Karon MD: Device-independent corneal topography classification and keratoconus grading by neural networks. ARVO Abstract #2867, Association for Research in Vision and Ophthalmology, Fort Lauderdale, Florida, April 25-30, 2004. [for text, see http://www.arvo.org]

223. Karon MD, Klyce SD, Smolek MK: Device-independent statistical indexes in corneal topography. ARVO Abstract #2874, Association for Research in Vision and Ophthalmology, Fort Lauderdale, Florida, April 25-30, 2004. [for text, see http://www.arvo.org]

224. Velasquez LM, Klyce SD, Karon MD, Courville CB: Mediation of Stromal Swelling Following Corneal Epithelial Trauma. ARVO Abstract #127, Association for Research in Vision and Ophthalmology, Fort Lauderdale, Florida, April 25-30, 2004. [for text, see http://www.arvo.org]

225. Klyce SD, Smolek MK: Device-independent classification of corneal topography with a parallel neural network approach. Abstracts from the Ninth Nidek International Refractive Surgery Symposium. June 19-19, 2004, Barcelona, Spain, p S755.

226. Klyce SD, Smolek MK: Interpreting wavefront. Abstracts from the Ninth Nidek International Refractive Surgery Symposium. June 19-19, 2004, Barcelona, Spain, p S756.

227. Alchagirov A, Klyce SD, Smolek MK, Karon, MD: Comparison of accuracy of zernike polynomials and fourier series in corneal surface representation. ARVO Abstract #851, Association for Research in Vision and Ophthalmology, Fort Lauderdale, Florida, May 1-5, 2005. [for text, see http://www.arvo.org]

228. Klyce SD: Zernike vs. Fourier Analysis for guiding individualized laser sculpting. Abstracts from the Tenth Nidek International Refractive Surgery Symposium. April 6, 2005, Dubai, UAE, p 17; J Refract. Surg. 2005;21:S664.

229. Klyce SD: Determination of the functional optical zone after keratorefractive surgery. Abstracts from the Tenth Nidek International Refractive Surgery Symposium. April 6, 2005, Dubai, UAE, p 18; J Refract. Surg. 2005;21:S664.

230. Klyce SD: Wavefront for dummies. Abstracts from the Tenth Nidek International Refractive Surgery Symposium. April 6, 2005, Dubai, UAE, p 18; J Refract. Surg. 2005;21:S664.

231. Klyce SD: Screening patients with the corneal navigator. Abstracts from the Tenth Nidek International Refractive Surgery Symposium. April 6, 2005, Dubai, UAE, p 18.

232. Klyce SD: Integrating ocular wavefront with corneal topography. Abstracts from the International VIIIth Congress of the Pan Arab African Council of Ophthalmology (PAACO) and the XIIth UAE Ophthalmic Conference April 7-10, 2005, Dubai, UAE, p 80.

233. Klyce SD, Smolek MK: Corneal topography and wavefront in clinical practice. Keynote Lecture. ISRS/AAO 2005 Emerging Trends in Refractive and Cataract Surgery. Hong Kong, China. May 14-16, p. 21.

234. Smolek MK, Klyce SD, Sarver EJ, Tan C: Corneal Ectasia Screening by Functional Optical Zone Area Analysis. Invest Ophthalmol Vis Sci. 2006;47:ARVO E-Abstract 568 [for text, see http://www.arvo.org]

235. Klyce SD, Tabernero J, Sarver EJ, Artal P: The Functional Optical Zone (FOZ) After Keratorefractive Surgery. Invest Ophthalmol Vis Sci. 2006;47:ARVO E-Abstract 2956. [for text, see http://www.arvo.org]

236. Klyce SD: Using the corneal confocal microscope in refractive surgery. Società Oftalmologica Italiana 4th International Congress. May 17-20, 2006, Rome, Italy, p 38-39.

237. Klyce SD: Screening for keratoconus in refractive surgery practices. Società Oftalmologica Italiana 4th International Congress. May 17-20, 2006, Rome, Italy, p 39-40.

238. Klyce SD: Corneal Topography 101. ISRS/AAO 2006 International Refractive Surgery: Art and Science. Istanbul, Turkey. May 26-28, 2006, p. 24.

239. Klyce SD: Zernike vs. Fourier. ISRS/AAO 2006 International Refractive Surgery: Art and Science. Istanbul, Turkey. May 26-28, 2006, pp. 58-59.

240. Klyce SD: Corneal refractive optics with topography and wavefront analysis. Abstracts of the XVII International Congress of Eye Research. Oct 29-Nov 3, 2006. Buenos Aires, Argentina.

241. Klyce SD: Understanding and applying wavefront analysis in your practice. Abstracts of the ISRS/AAO Meeting, May 25-27, 2007, Beijing, China, p 8.

242. Klyce SD: Effective patient screening for forme fruste keratoconus with topography and pachymetry. Abstracts of the ISRS/AAO Meeting, May 25-27, 2007, Beijing, China, p 10.

243. Klyce SD: Corneal confocal microscopy applications in refractive surgery. Abstracts of the ISRS/AAO Meeting, May 25-27, 2007, Beijing, China, p 20.

244. Klyce SD: Clinical pearls for effective use of corneal topography in your practice. Abstracts of the XXVII Pan-American Congress of Ophthalmology, May 29-June 3, 2007, Cancun, Mexico, p 212.

245. Smolek MK, Vujosevic S, Piermarocchi S, Midena E, Peto T, De Luca M, Grisan E, Ruggeri A, Klyce SD: Investigational use of an automated diabetic retinopathy screening with an artificially-intelligent, non-mydriatic, operator-free fundus camera. American Diabetes Association Annual Meeting, 2008

246. Smolek MK, Vujosevic S, Piermarocchi S, Midena E, Peto T, De Luca M, Grisan E, Ruggeri A, and Klyce SD. An expert system for diabetic retinopathy screening with a non-mydriatic, operator-free fundus camera. ARVO Meeting Abstracts April 11, 2008 49:2725.

247. Roberts CJ, Mahmoud AM, Liu J, Sharalaya Z, Mauger TF, Lembach RG, Hendershot AJ, Kuennen R, Klyce SD. Conservation of Arclength in Keratoconic and Normal Corneas with Air Puff Induced Deformation. ARVO Meeting Abstracts March 26, 2012 53:6893.

248. Klyce SD, van den Berg RM, van den Berg A, Dodhia M, Shahid M, Rocha KM. Estimation of Visual Benefit with IC-8 in Keratoconus Patients. ASCRS Meeting Poster, 2024.

249. de Barros, M, Lottelli L, Rocha KM, van den Berg R, Bedi R, Klyce SD. Subjective versus objective depth of focus from ray-tracing aberrometry in small aperture optics intraocular lens in eyes with aberrated corneas. ASCRS Meeting Abstract, 2024.

250. DeVaro S, Barros M, van den Berg RM, van den Berg AB, Bedi R, Klyce SD, Rocha KM. Estimating the effectiveness of small aperture IOL in reducing higher order aberrations in post-radial keratotomy patients. ASCRS Meeting Abstract, 2024.

251. de Barros MF, Rocha KM, van den Berg R, Klyce SD. Comparison of subjective and objective depth of focus curve using ray-tracing aberrometry in aberrated corneas implanted with a small aperture intraocular lens. ESCRS Meeting Poster, 2024.

252. Rocha K, Barros M, van den Berg R, Klyce SD. Visual performance and objective depth of focus from ray tracing aberrometry in small aperture optics intraocular lens in eyes with aberrated corneas. AOS 161st Annual Meeting Abstracts. May 15, 2025, p 23.

253. Klyce S. AI in corneal diagnostics: where are we after 3 decades? Aegean Cornea XVII & Cataract Abstracts. June 26-29, 2025; Crete, Greece.

254. Klyce S, Barros M. Visual outcomes and depth of focus from ray tracing aberrometry in small aperture optics intraocular lens in eyes with aberrated corneas. ESCRS. September 16, 2025, Copenhagen, Denmark.

EDITORIAL ACTIVITIES

1. Klyce SD: Open up the VAXBI. Letter to the Editor. *Digital Review*, October 13, 1986, p. 50.

2. Dingeldein SA, Klyce SD: The effect of suture removal on postkeratoplasty astigmatism. *Am J Ophthalmol* 106:507, 1988.

3. Wilson SE, Klyce SD: Topographic analysis and visual acuity after radial keratotomy. *Am J Ophthalmol* 107:436-437, 1989.

4. McDonald MB, Kaufman HE, Frantz JM, Shofner S, Salmeron B, Klyce SD: Excimer laser ablation in a human eye. Case report. *Arch Ophthalmol* 107:641-642, 1989.

5. Klyce SD, Wilson SE, Kaufman HE: Corneal topography comes of age. *Refractive & Corneal Surgery* 5:359-361, 1989.

6. Klyce SD, Wilson SE: Editors, Special Issue on Corneal Topography. *Refractive & Corneal Surgery* 5: November/December 1989.

7. Limberg MB, Dingeldein SA, Green MT, Klyce SD, Insler MS, Kaufman HE. Corneal Compression Sutures for the Reduction of Astigmatism After Penetrating Keratoplasty: Reply American Journal of Ophthalmology, Volume 108, Issue 5, November 1989, Page 614.

8. Klyce SD: How to make a success of your ophthalmic research programme. *Asia-Pacific Journal of Ophthalmology* 2:27-28, 1990.

9. Rabinowitz YS, Klyce SD, Krachmer J, et al: Opinions. Keratoconus, videokeratography, and refractive surgery. *Refractive & Corneal Surgery* 8:403-407, 1992.

10. Klyce SD: In Opinions. Nomenclature for keratoconus suspects. *Refractive & Corneal Surgery* 9:220-221, 1993.

11. Klyce SD: Computed Anatomy TMS-1 Videokeratoscope. *In Focus* 1:2, 1993.

12. Maeda N, Hosotani H, Miyata K, Sakimoto T, Klyce SD, Hamano H: Clinical methods for evaluating corneal topography. *J Japanese Contact Lens Soc* 36:62-67, 1994.

13. Ribeiro JC, McDonald MB, Klyce SD: Photorefractive keratectomy after radial keratotomy in a patient with severe myopia. *Am J Ophthalmol* 118:106-108, 1994

14. Maeda N, Klyce SD, Smolek MK: RK on keratoconus: a question of safety. *J Cataract Refract Surg* 20:670, 1994

15. Klyce SD: Preface to Atlante di Topografia Corneale by P Vinciguerra. Fogliazza editore, Milano, 1995, p. XIII.

16. Klyce SD: Information fidelity in corneal topography. [Invited editorial] *Br J Ophthalmol* 79: 791-792, 1995

17. Klyce SD, Smolek MK, Maeda N: [Letter to the Editor, reply] Goren MB: [Letter to the editor] "Comparison of methods for detecting keratoconus using videokeratography." (Maeda N, Klyce SD, Smolek MK, In July 1995 Issue; *Arch Ophthalmol*) *Arch Ophthalmol* 114:631-632, 1996

18. Klyce SD, Jacob JT: [Letter to the editor] PRK and collagen denaturation, *Cornea* 16:598, 1997.

19. Klyce SD, Smolek MK, Maeda N: Keratoconus detection with the KISA% method - another view. *J Cataract Refract Surg*, 26:472-473, 2000.

20. Ambrósio R Jr, Klyce SD, Smolek MK, Wilson SE: Pellucid marginal corneal degeneration. *J Refract Surg* 18:86-88, 2002.

21. Klyce SD: Night vision after LASIK: the pupil proclaims innocence. [Invited] Guest Editorial. *Ophthalmology*. 2004;111:1-2.

22. Smolek MK, Klyce SD: Author Response: Inadequacy of Zernike Polynomial Fitting: Case Not Made. *Invest Ophthal Vis Sci* eLetter. 27 Feb 2004. [for text, see http://www.arvo.org]

23. Klyce SD: Evaluating aberrations induced after refractive surgery must take pupil size into account. Letter to the Editor. *J Refract Surg.* 2005;21:764-5.

24. Klyce SD: Night vision disturbances after refractive surgery: haloes are not just for angels. [Invited] Guest Editorial. *Br J Ophthalmol,* 2007;91:992-3.

25. Klyce SD: Foreword to *Keratoconus Surgery and Cross-Linking.* Ed., Pinelli R, Leccisotti A, Jaypee Brothers Medical Publishers, Ltd, Maryland Heights, MO, p. IX, 2009.

26. Klyce SD: Chasing the suspect: keratoconus. [Invited] Guest Editorial. *Br J Ophthalmol,* 2009;93:845-7.

27. Klyce SD: Can you spot the keratoconus suspect? [Invited] Editorial. *Int J Keratoco Ectatic Corneal Dis*, 2012;1(3):v-vi.

28. Klyce SD: Research Highlight. UVA-riboflavin collagen cross-linking (CXL)—a Misnomer Perhaps, but it Works! *Invest Ophthalmol Vis Sci*. 2013;54(3):1635.

29. Klyce SD: Preface. In: *Corneal Topography*. Ed. Kiliç A, Roberts CJ. Kugler Publications. Amsterdam, The Netherlands, 2013.

30. Randleman JB, Dupps WJ Jr, Santhiago MR, Rabinowitz YS, Koch DD, Stulting RD, Klyce SD. Screening for Keratoconus and Related Ectatic Corneal Disorders. *Cornea*. 2015;34:e20-2.

31. Klyce SD: The future of keratoconus screening with artificial intelligence. [Invited] Commentary. *Ophthalmology* 2018;125(12), 1872-3.

32. Klyce SD: Introduction. In: *Atlas of Corneal Imaging.* Ed. Randleman JB, Santhiago MR, Dupps WJ Jr. Slack, Inc., Thorofare, NJ. 2022.

PROCEEDINGS, TRANSACTIONS, NON-PEER REVIEWED ARTICLES

1. Vision Research. A National Plan. Report of the Corneal Diseases Panel. National Institutes of Health Pub. No. 83-2472. [Contributor to chapters] 2. Ocular Surface Problems; 3. Refractive Problems and Contact Lenses; 4. Corneal Edema, Endothelial Dysfunction, Dystrophies, and Inherited Disease. 1983

2. Klyce SD, et al: Panel Discussion. Effects of daily and extended wear contact lenses on the cornea. I. Epithelium and stroma. In: *Transactions of the International Society for Contact Lens Research*, Third Scientific Meeting, Cambridge, September 23-25, 1984. Ed., Kenyon E, Kensington, I.S.C.L.R., 1986, pp. 63-90

3. Klyce SD, et al: Panel Discussion. Effects of daily and extended wear contact lenses on the cornea. II. Stroma and endothelium. In: *Transactions of the International Society for Contact Lens Research*, Third Scientific Meeting, Cambridge, September 23-25, 1984. Ed., Kenyon E, Kensington, I.S.C.L.R., 1986, pp. 91-124

4. Klyce SD, et al: Panel discussion. Anatomy and Physiology. In: *Transactions of the International Society for Contact Lens Research*. 4th Scientific Meeting, S'Agaro, Spain, Sept. 15-18, 1986. Ed. Guillon M, Polse KA, pp. 43-83.

5. Klyce SD, et al: Panel Discussion. Refractive Surgery. In: *Transactions of the International Society for Contact Lens Research*. 4th Scientific Meeting, S'Agaro, Spain, Sept. 15-18, 1986. Ed. Guillon M, Polse KA, pp.85-126.

6. Klyce SD, et al: Panel Discussion. Pharmacology, Ocular Toxicology and Immunology. In: *Transactions of the International Society for Contact Lens Research*. 4th Scientific Meeting, S'Agaro, Spain, Sept. 15-18, 1986. Ed. Guillon M, Polse KA, pp. 239-280.

7. Klyce SD et al: Panel Discussion. Research Techniques and Methods: Current Frontiers. In: *Transactions of the International Society for Contact Lens Research*. 4th Scientific Meeting, Poipu Beach, August 31-Sept. 3, 1988, pp. 288-329.

8. Klyce SD, Wilson SE: Imaging, reconstruction, and display of corneal topography. *Proceedings of the SPIE 33rd Annual International Technical Symposium of Optical and Optoelectronic Applied Science and Engineering*, New methods in microscopy and low light imaging. *SPIE* 1161:409-416, 1989.

9. Klyce SD: Modern corneal topography analysis. *6th Scientific Research Conference Proceedings of the International Society for Contact Lens Research*. Monte Carlo, August 31-September 3, 1990, pp. 55-56.

10. Klyce SD: Curvature and topography. In: *Corneal Biophysics Workshop II. Optics, measurement, and modeling of the cornea*. Eds McNicol LA, Strahlman E, National Eye Institute, 1990, 35-52.

11. Klyce SD: Current state of empirical and heuristic models. In: *Corneal Biophysics Workshop II. Optics, measurement, and modeling of the cornea*. Eds McNicol LA, Strahlman E, National Eye Institute, 1990, 209-214.

12. Klyce SD: Corneal topography and contact lenses. *Considerations in Contact Lens Use Under Adverse Conditions*, National Research Council, Committee on Vision, National Academy Press, Washington D.C. 1991; pp 68-73.

13. Klyce SD: Corneal topography in spherical defects. In: *Cataract and Refractive Microsurgery. Proceedings of the V International Congres*, Eds Maumenee AE, Stark WJ, Isente I, Bologna, Italy, Cappelli Editore, 1991, pp. 225-228.

14. McDonald MD, Klyce SD: Refractive surgery with the excimer laser. Part I. *Highlights of Ophthalmology* XIX, No.6, 1991, pp. 35-42.

15. Klyce SD: Clinical applications of corneal topography. In: *Research to Prevent Blindness Science Writers Seminar*, Universal City, CA, April 25-28, 1993, pp. 62-63.

16. Klyce SD: Corneal topography: why the refractive cataract surgeon won't be able to live without it. In: Small Incision and Refractive Cataract Surgery. Proceedings of the Second Annual Ocular Surgery News Symposium, *Supplement to Ocular Surgery News*, Feb 15, 1994, p 66.

17. Klyce SD: Corneal shape pre and post rk and prk and their optical implications. *7th Scientific Research Conference Proceedings of the International Society for Contact Lens Research.* Hayman Island, September 6-11, 1993.

18. Klyce SD: Fitting contact lenses after refractive surgery and corneal transplantation. *7th Scientific Research Conference Proceedings of the International Society for Contact Lens Research.* Hayman Island, September 6-11, 1993.

19. Maeda N, Klyce SD, Smolek MK: Automated keratoconus detection for refractive surgery and contact lens candidates with corneal topography. *7th Scientific Research Conference Proceedings of the International Society for Contact Lens Research.* Hayman Island, September 6-11, 1993.

20. Klyce, SD: A comparison of irregular astigmatism comparing RK, PRK, ALK, and LASIK. *Proceedings from the Fourth Annual Ocular Surgery News Symposium. Cataract and Refractive Surgery.* New York, September 16-17, 1995, p 40.

21. Klyce SD: Comparison of topographic methods for assessing corneal optics in refractive surgery. *Proceedings from Mopane '96. Refraction and Keratometry: the Mathematics and Statistics.* Kruger Park, South Africa, August 3-5, 1996, pp 133-138.

22. Klyce, SD: Measuring potential optical performance with corneal topography after refractive surgery. *Proceedings from the Fifth Annual Ocular Surgery News Symposium. Cataract and Refractive Surgery.* New York, October 5-6, 1996, p 131-133.

23. Klyce, SD: Effective screening for ectasia risk. *Refractive Eyecare* March 2012, 7-8.

24. Klyce, SD: Corneal physiology and topography in relation to excimer laser surgery. *Proceedings of the Philippine Academy of Ophthalmology.* Manila, March 8-12, 1999.Klyce, SD: The Two Sides to Measuring Corneal Astigmatism. *Cat. Ref. Surg. Today.* May 2014, 1-2.

25. Klyce, SD: Getting the most from topography. *Cat. Ref. Surg. Today.* July 2014, 1-2.

26. Klyce, SD: The Unique Optical System of the LENSAR Laser System. The technology behind Augmented Reality imaging and how it benefits laser cataract surgery. *Cat. Ref. Surg. Today.* Insert. July 2014, 1-2.

27. Klyce, SD: Exploring the utility of the OPD-SCAN III to detect keratoconus and corneal ectasias. *Cat. Ref. Surg. Today, Europe Suppl.* Sept 2021, 34-35.

28. Klyce SD, McDonald MB: The future of corneal surgery. *Cat. Ref. Surg. Today* Jan 2024 24(1):1.

29. Klyce SD, McDonald MB: A new approach to thermal keratoplasty. Will it finally work? *Cat. Ref. Surg. Today* Jan 2024 24(1):51-52,64.

BOOK CHAPTERS

1. Klyce SD: Cl transport in rabbit cornea. In: *Cl Transport in Biological Membranes*. American Physiological Society Symposium. Ed., Zadunaisky JA, New York, Academic Press, 199-221, 1982

2. Wagner NJ, Kaufman HE, Beuerman RW, Klyce SD: Physiology of the cornea. Chapter 5. In: *Contact Lenses, The CLAO Guide to Basic Science and Clinical Practice*. Ed., Dabezies OH, Jr. Orlando, Grune and Stratton, 1984, pp 5.1-5.13

3. Rosenthal P, Klyce SD: Contact lens induced corneal edema. Chapter 15. In: *Contact Lenses, The CLAO Guide to Basic Science and Clinical Practice*. Ed., Dabezies OH. Orlando, Grune and Stratton, 1984, pp 15.1-15.11

4. Klyce SD, Farris RL, Dabezies OH: Corneal oxygenation in contact lens wearers. Chapter 14. In: *Contact Lenses, The CLAO Guide to Basic Science and Clinical Practice*. Ed., Dabezies OH, Jr., Orlando, Grune and Stratton, 1984, pp 14.1-14.5

5. Marshall WS, Klyce SD: Control mechanisms of Cl^- secretion in some vertebrate epithelia. In: *Proceedings IX International Symposium on Comparative Endocrinology*, Current Trends in Comparative Endocrinology, Eds., Lofts B and Holmes WN, Hong Kong University Press, 1:971-974, 1985

6. Klyce SD, McCarey BE: Physiology of the cornea. In: *Contact Lenses, The CLAO Guide to Basic Science and Clinical Practice*. Ed., Dabezies OH, Jr., Orlando, Grune and Stratton, 1986, pp. 5.1-5.14

7. Klyce SD: Computer analysis of corneal topography. In: *Corneal Surgery: Theory, Technique, and Tissue*. Ed., FS Brightbill. St. Louis, C.V. Mosby Co., 1986, pp. 563-568

8. Klyce SD, Bonanno JA: Role of the epithelium in corneal hydration. In: *The Cornea: Transactions of the World Congress on the Cornea III*. Eds., HD Cavanagh, MA Lemp. New York, Raven Press, 1988, pp. 159-164

9. Klyce SD, Beuerman RW: Anatomy and physiology of the cornea. In: *The Cornea*. Ed., Kaufman HE, Barron BA, McDonald MB, Waltman SR. New York, Churchill Livingstone Inc., 1988, 3-54.

10. Lin DTC, Wilson SE, Klyce SD: Alteration of corneal topography by refractive keratoplasty. In: *Repair Processes in the Cornea*. Eds., Beuerman RW, Crosson CE, Kaufman HE. The Woodlands, Portfolio Publishing Company of Texas, Inc., 1989, 183-193.

11. Klyce SD, Dingeldein SA: Corneal topography. In: *Noninvasive diagnostic techniques in ophthalmology*. Ed., Masters BR. New York, Springer-Verlag Inc., 1990, 61-77.

12. Klyce SD, Wang J: Considerations in corneal surface reconstruction from keratoscope images. In: *Noninvasive diagnostic techniques in ophthalmology*. Ed., Masters BR. New York, Springer-Verlag Inc., 1990, 78-81.

13. Wilson SE, Wang J, Klyce SD: Quantification and mathematical analysis of photokeratoscopic images. In: *Corneal Topography*. *Measuring and Modifying the Cornea.* Eds., Schanzlin DJ and Robin J. New York, Springer-Verlag Inc., 1992, 1-9.

14. Constad WH, Klyce SD: Corneal topography and hard contact lenses. In: *Contact Lenses, The CLAO Guide to Basic Science and Clinical Practice*. Second edition, Update 9, Ed., Dabezies OH, Jr., Boston, Little, Brown, & Co, 1992, 4A.1-16

15. Klyce SD: Corneal topography in refractive keratectomy. In: *Color Atlas/Text of Excimer Laser Surgery. The Cornea*. Eds., Thompson FB, McDonnell PJ., New York, Igaku-Shoin Medical Publishers, Inc, 1993, 19-36.

16. Smolek MK, Klyce SD: Cornea. In: *Duane's Foundations of Clinical Ophthalmology*, Chapter 8, Tasman W, Jaeger EA, (Eds), JB Lippincott, Philadelphia, 1993, 1-20.

17. Klyce SD, McDonald MB: Computerized Corneal Topography of Surface Ablations with the Tomey TMS-1. In: *Corneal Laser Surgery.* Eds Salz JJ, McDonnell PJ, McDonald MB, Mosby, St Louis, 1995, 93-108.

18. Smolek MK, Klyce SD: Physiology of the cornea. In: *Contact Lenses, The CLAO Guide to Basic Science and Clinical Practice. Volume 1: Basic Science.* Ed., Kastl, PR., Kendall/Hunt, Dubuque, 1995, 19-40.

19. Klyce SD: Corneal oxygenation and edema. In: *Contact Lenses, The CLAO Guide to Basic Science and Clinical Practice. Volume 1: Basic Science.* Ed., Kastl, PR., Kendall/Hunt, Dubuque, 1995, 205-220.

20. Smolek MK, Klyce SD: The Tomey/Computed Anatomy TMS-1 Videokeratoscope. In *Corneal Topography. The State of the Art.* Eds Gills JP, Sanders DR, Thornton SP, Martin RG, Gayton JL, Holladay JT. Slack, Thorofare, NJ, 1995, pp 123-149.

21. Klyce SD: Corneal mapping. In: *Atlas of Corneal Topography.* Eds: Buratto L, Cantera E, Dal Fiume E, Genisi C, Merlin U. CAMO srl, Milano, 1995, 13-16.

22. Mammone RJ, Klyce SD, Zhang X, Gersten M: Computer-assisted eye surgery. Chapter 1. In: *Computer-Integrated Surgery. Technology and Clinical Applications.* Eds. Taylor RH, Lavallee S, Burdea GC, Mosges R. Cambridge, MIT Press, 1995, 473-481.

23. Klyce SD: Role of corneal topography in refractive surgery. In: *Cornea.* Eds Krachmer JH, Mannis MJ, Holland EJ. New York, Mosby, 1996, 2039-2058.

24. Klyce SD: Topographic comparison of current refractive surgical modalities. In: *Lamellar Refractive Surgery*. Casebeer JC, Ruiz LA, Slade S. Slack, Inc., Thorofare, N.J., 1996.

25. Klyce SD: Corneal topography. Clinical applications. In: *Principles and Practice of Refractive Surgery.* Eds Elander R, Rich LF, Robin JB. WB Saunders, Philadelphia, 1997, 499-512.

26. Beuerman RW, Klyce SD, Kaufman HE: Functional anatomy of the cornea. In: *Corneal Physiology and Disposable Contact Lenses.* Hamano H, Kaufman HE. Boston, Butterworth-Heinemann, 1997, 3-16.

27. Klyce SD, Maeda N: Corneal topography. In: *Corneal Physiology and Disposable Contact Lenses.* Hamano H, Kaufman HE. Boston, Butterworth-Heinemann, 1997, 43-62.

28. Smolek MK, Klyce SD: The principles of computerized videokeratoscopy. In: *Excimer Lasers in Ophthalmology; Principles and Practice*. McGhee CNJ, Taylor HR, Gartry DS, Trokel SL (eds). London, Martin Dunitz, Ltd. 1997; Chapter 6, pp.79-96.

29. Martinez CE, Klyce SD, Waring, GO, III, El Maghraby A: The topography of LASIK. Chapter 30. In: *LASIK*. Eds Pallikaris IG, Siganos DS. Slack, Inc, Thorofare, NJ. 1997, pp 339-357.

30. Klyce SD, Beuerman RW: Anatomy and physiology of the cornea. Chapter 1. In: *The Cornea*. 2nd edition. Eds Kaufman HE, Barron BA, McDonald MB. Boston, Butterworth-Heinemann, 1998, 3-50.

31. Dabezies OH, Klyce SD, Morgan JF, Hartstein J, Donshik PC, Boswall GJ, Ehlers WH, Stein RM, Doughman DJ: Corneal changes from contact lenses. Chapter 27. In: *The Cornea*. 2nd edition. Eds Kaufman HE, Barron BA, McDonald MB. Boston, Butterworth-Heinemann, 1998, 697-714.

32. Klyce SD, Maeda N, Byrd TJ: Corneal topography. Chapter 46. In: *The Cornea*. 2nd edition. Eds Kaufman HE, Barron BA, McDonald MB. Boston, Butterworth-Heinemann, 1998, 1055-1075.

33. Wilson SE, Klyce SD: "Advances in Computerized Corneal Topographic Analysis? Where's the Beef?" In: Long DA (ed.): Cornea and Refractive Surgery, New Orleans Academy of Ophthalmology. Copyright © 1998 Kugler Publications, Hague, The Netherlands.

34. Wilson SE, Klyce SD: Background of corneal topography. In: *Refractive Surgery*. Eds Wu HK, Thompson VM, Steinert RF, Slade SG, Hersh PS. Thieme, New York, NY, 1999, 53-62.

35. Oshika T, Klyce SD: Corneal topography: basic concepts. In: *Corneal Surgery: Theory, Technique, & Tissue. Part II. Testing and Measuring Corneal Function prior to Surgery.* 3rd edition. Ed., FS Brightbill. Mosby, Inc., St. Louis, 1999, 91-97, Color Plates 4-13.

36. Dohlman CH, Klyce SD: Corneal edema. Chapter 59. In: *Principles and Practice of Ophthalmology*. Albert DM, Jakobiec F. W.B. Saunders, Philadelphia, 2000, 646-657.

37. Klyce SD, Martinez CE: Corneal topography. Chapter 61. In: *Principles and Practice of Ophthalmology*. Albert DM, Jakobiec F. W.B. Saunders, Philadelphia, 2000, 668-694.

38. Wilson SE, Walker M, Klyce SD: LASIK decentration and central islands. Chapter 34. In: *LASIK. Surgical Techniques and Complications.* Ed Buratto L and Brint S. Slack, Inc, Thorofare, NJ. 2000, pp 565-580.

39. Klyce SD: Corneal topography. Chapter 5. In: *The Cornea in Contact Lens Practice*. Ed Efron N. Butterworth-Heinemann, Oxford, UK, 2001, pp 178-199.

40. Applegate RA, Howland HC, Klyce SD: Corneal aberrations and refractive surgery. Chapter 20. In: Customized Refractive Surgery - Super Vision. Eds MacRae S, Krueger R, Applegate RA. Slack, Inc, Thorofare, NJ, 2001, pp 239-246.

41. Endl MJ, Klyce SD: Surgical planning with videokeratography. Chapter 1.1 In: *101 Pearls in Refractive, Cataract, and Corneal Surgery*. Eds Melki SA, Azar DT. Slack, Inc. Thorofare, NJ, 2001, pp 3-8.

42. Ruberti, JW, Klyce SD. Physiological System Models of the Cornea. In *Models of the Visual System.* Eds: Hung, GK, Ciuffreda KJ. Topics in Biomedical Engineering International Book Series. Ed: Micheli-Tzanakou, E. Kluwer Academic/Plenum Publishers, New York, NY, 2002, pp 3-55.

43. Klyce SD, Endl MJ: Corneal topography in modern refractive surgery. Chapter 2. In Frontiers in Refractive Surgery. Ed Hannush SB. Lippincott, Williams, and Wilkins, Philadelphia, PA. Intl Ophthal Clinics 42:19-30, 2002.

44. Smolek MK, Klyce SD: Basic optics of hyperopia and presbyopia. Chapter 2. In *Hyperopia and Presbyopia*. Ed Azar D, Koch D. Marcel Dekker, New York, NY, 2003, pp 17-26.

45. Klyce SD, Smolek MK, Endl MJ, Malineni V, Insler MS, McDonald MB: Topographic changes. Chapter 13. In *Hyperopia and Presbyopia*. Ed Azar D, Koch D. Marcel Dekker, New York, NY, 2003, pp 129-140.

46. Klyce SD: Corneal topography of keratoconus. Chapter 8. In *Kératocône et lentilles de contact.* Ed Malet F. Imprimerie LAMY, Marseille, France, 2003, pp 83-92 [English]; pp 93-98 [Francais].

47. Klyce SD: Corneal physiology. In Smolin and Thoft's *The Cornea. Scientific foundations and clinical practice*, IVth Edition, Eds CS Foster, DT Azar, CH Dohlman. Lippincott, Williams, and Wilkins, New York, 2004, pp 37-58.

48. Courville CB, Klyce SD: Corneal topography. In Smolin and Thoft's *The Cornea. Scientific foundations and clinical practice*, IVth Edition, Eds CS Foster, DT Azar, CH Dohlman. Lippincott, Williams, and Wilkins, New York, 2004, pp 175-186.

49. Smolek MK, Klyce SD: Keratometry. In Smolin and Thoft's *The Cornea. Scientific foundations and clinical practice*, IVth Edition, Eds CS Foster, DT Azar, CH Dohlman. Lippincott, Williams, and Wilkins, New York, 2004, pp 169-174.

50. Klyce SD: Wavefront analysis. Section 1. In: *Focal Points. Clinical Modules for Ophthalmologists*. 23:10. Ed by GA Stern. American Academy of Ophthalmology, San Francisco, CA, 2005, pp 1-10.

51. Klyce SD, Endl MJ: Surgical planning with videokeratography. Chapter 1. In: *101 Pearls in Refractive, Cataract, and Corneal Surgery*, 2nd Edition. Eds Melki SA, Azar DT. Slack, Inc. Thorofare, NJ, 2006, pp 1-16.

52. Klyce SD: The Tomey TMS corneal topographer. In: *Corneal Topography in the Wavefront Era*, Chapter 22, Ed by Wang M, Slack, Inc, Thorofare, NJ, 2006, pp 249-58.

53. Klyce SD: Understanding wavefront: microns versus diopters. In: *Refractive Surface Ablation. PRK, LASEK, Epi-LASIK, Custom, PTK, and Retreatment,* Chapter 4, Ed by Vinciguerra P, Camesasca F, Slack, Inc, Thorofare, NJ, 2006, pp 33-39.

54. Smolek MK, Klyce SD: Cornea. In: *Duane's Foundations of Clinical Ophthalmology*, Chapter 8, Tasman W, Jaeger EA, (Eds), JB Lippincott, Philadelphia, 2007, 47 pp, online edition.

55. Klyce SD, Oshika T: Placido-based topography. In: *Corneal Surgery: Theory, Technique, & Tissue. Part II. Testing and Measuring Corneal Function.* 4th edition. Ed., FS Brightbill FS, McDonnell PJ, Farjo AA, McGhee CNJ, Serdarevic ON, Mosby Elsevier, New York, NY, 2009, pp 75-82.

56. Klyce SD, Dohlman CH, Martinez C: Corneal Physiology. In: *Principles and Practice of Ophthalmology*. Albert DM, Jakobiec F. 3rd Edition, In press.

57. Endl MJ, Klyce SD: Detecting corneal ectasia. In: *Keratoconus Surgery and Cross-Linking.* Ed., Pinelli R, Leccisotti A, Jaypee Brothers Medical Publishers, Ltd, Maryland Heights, MO, 2009, pp 1-12.

58. Klyce SD: Topography in the diagnosis of keratoconus and keratoectasia. In *Keratoconus and Keratoectasia: Prevention, Diagnosis, and Treatment*. Ed., M Wang, T Swarz. Slack, Inc., Thorofare, N.J., 2009, 61-7.

59. Martinez CE, Klyce SD: Keratometry and Topography. In: *Cornea*. Section 3. Imaging Techniques of the Cornea. Chapter 13. Ed. Krachmer JH, Mark J Mannis MJ, Holland EJ. 3rd Edition, 2011, 161-176.

60. Klyce SD: Corneal Topography: Screening and Identifying FFKC. In: *Curbside Consultation in Refractive Surgery*, Ed. Donnenfeld E, McDonald M, Solomon K, in press

61. Klyce SD, Tannen BL: The Tomey TMS corneal topographer. In: *Corneal Topography in the Wavefront Era*, Chapter 22, Ed by Wang M, Swartz T, 2nd Edition, Slack, Inc, Thorofare, NJ, 2012, 51-56.

62. Smolek MK, Klyce SD: Cornea. In: *Duane's Ophthalmology*, Chapter 8, Tasman W, Jaeger EA, (Eds), JB Lippincott, Philadelphia, 2012, 47 pp, online edition.

63. Klyce SD, Palanker DV, Edwards KH, Krueger RR: Imaging Systems and Image-guided Surgery. In: *Textbook of Refractive Laser Assisted Cataract Surgery (ReLACS).* Chapter 5, Ed by Krueger RR, Talamo JH, Lindstrom RL. Springer, New York, NY, 2013, pp 49-58.

64. Waring GO, IV, Klyce SD: Intrastromal Corneal Implants for the Treatment of Presbyopia. In: *Copeland and Afshari's Principles and Practice of Cornea,* Volume 2, Section 16, Chapter 99, Ed by Copeland Jr RA, Afshari NA. Jaypee Brothers Medical Publishers, Ltd, New Delhi, 2013, pp 1274-9.

65. Klyce SD, Espana EM, Waring GO IV. Preoperative Considerations: Corneal Topography. In: *Copeland and Afshari's Principles and Practice of Cornea.* Volume 2, Section 16, Chapter 94. Ed by Copeland Jr RA, Afshari NA. Jaypee Brothers Medical Publishers, Ltd, New Delhi, 2013, pp 1213-30.

66. Kim EJ, Weikert MP, Martinez CE, Klyce SD. Keratometry and Topography. In: *Cornea*. Volume 1. Chapter 12. Ed by Mannis MJ. Holland EJ. Reed Elsevier India, Chennai, 2017, pp 144-53.

67. Klyce S, Hallak J, Romond K, Azar D, Kim T. (2021) Corneal Physiology: Corneal Form and Function. In: Albert D., Miller J., Azar D., Young L.H. (eds) Albert and Jakobiec's Principles and Practice of Ophthalmology. Springer, Cham. https://doi.org/10.1007/978-3-319-90495-5_203-1

BOOK

1. Rabinowitz YS, Wilson SE, Klyce SD: Atlas of Corneal Topography, New York, Igaku-Shoin Medical Publishers, Inc, 1993.

NATIONAL AND INTERNATIONAL STANDARDS

1. Klyce SD. Contributor, American National Standard Institute. Ophthalmics—Corneal Topography Systems—Standard Terminology. Requirements. ANSI Z80.23–1999. Optical Laboratories Association, American National Standards Institute, Inc, 1999.

2. Klyce SD. Contributor, American National Standards Institute. Laser systems for corneal reshaping. ANSI Z80.11-2007 (Optical Laboratories Association, Z80 Secretariat, Fairfax, VA 22030).

3. Klyce SD. Contributor, American National Standard Institute. Ophthalmics—Corneal Topography Systems—Standard Terminology. Requirements. ANSI Z80.23–2008. Optical Laboratories Association, American National Standards Institute, Inc. 2008.

4. Klyce SD. Contributor, Ophthalmic instruments - Corneal topographers. International Organization for Standardization (ISO). ISO 19980:2012.

5. Klyce SD. Major contributor, DICOM Working Group 9. Digital Imaging and Communications in Medicine. Supplement 168. Corneal Topography Map Storage SOP Class. 2013.

6. Klyce SD. Major contributor, DICOM Working Group 9. Digital Imaging and Communications in Medicine. Supplement 168. Corneal Topography Map Storage SOP Class. Revisions 2025 –

PATENTS

1. Klyce SD. System for computerized fitting of contact lenses. U.S. Patent #7,370,969. March 8, 1994.

2. Khoobehi B, Klyce SD, McDonald MB, Shaibani S: Laser surgical method of sculpting a patient's cornea and associated intermediate controlling mask. U.S. Patent #5,376,086. December 27, 1994.

3. Klyce SD, Smolek MK, Fujieda M: Corneal topography analysis system. U.S. Patent #7,370,969. May 13, 2008.

4. Wallace DA, Klyce SD, Rogers JR, Mulder RS, Kahan MA, Glenn PE. Mobile communication device-based corneal topography system | [système de topographie cornéenne basé sur un dispositif de communication mobile]. #WO2019165227. August 29, 2019.

5. Wallace D, Klyce SD, Rogers JR, Buscemi P, Kahan MA, Glenn PE, Tanassi C, Kramer D, Mehta V. Corneal topography system and methods [système et procédés de topographie cornéenne]. #WO2020205860 (A1). October 8, 2020.

6. Wallace DA, Klyce SD, Rogers JR, Mulder RS, Kahan MA, Glenn PE. Mobile Communication Device-Based Corneal Topography System. Japan Patent 2020-544793.

7. Wallace DA, Klyce SD, Rogers JR, Mulder RS, Kahan MA, Glenn PE. Mobile communication device-based corneal topography system | [auf mobiler kommunikationsvorrichtung basierendes hornhauttopografiesystem] European Patent # EP3755205 (A1). December 30, 2020.

8. Wallace D, Klyce SD, Rogers JR, Mulder RS, Kahan MA, Glenn PE. Corneal Topography Systems and Methods. U.S. Patent #11,096,573 August 4, 2021.

9. Wallace DA, Klyce S, Buscemi P, Kahan MA, Glenn PE, Rogers JR, Tanassi C, Kramer D, Mehta V. Corneal Topography System and Method of Operations. U.S. Patent #11,219,360 January 11, 2022.

10. Wallace DA, Klyce S, Rogers JR, Mulder RS, Kahan MA, Glenn PE. Corneal topography system and methods. U.S. Patent #11,219,361. January 11, 2022.

11. Wallace DA, Klyce S, Buscemi P, Kahan MA, Glenn PE, Rogers JR, Tanassi C, Kramer D, Mehta V. Corneal Topography System and Methods Including Illumination Assembly. U.S. Patent #11,224,341. January 18, 2022.

12. Wallace DA, Klyce S, Rogers JR, Mulder RS, Kahan MA, Glenn PE. Mobile Communication Device-Based Corneal Topography System; U.S. Patent #11,412,926. August 16, 2022.

13. Wallace DA, Klyce S, Buscemi P, Kahan MA, Glenn PE, Rogers JR, Tanassi C, Kramer D, Mehta V. Corneal topography system operations. U.S. Patent #11,471,046. October 18, 2022.

14. Wallace D, Klyce SD. Cloud Based Corneal Surface Difference Mapping System and Method. U.S. Patent #11,547,295. January 10, 2023.

15. Wallace D, Klyce SD. Cloud based corneal surface difference mapping system and method. U.S. Patent #11,576,573. February 14, 2023.

16. Wallace D, Klyce SD, Rogers JR, Mulder RS, Kahan MA, Glenn PE. Corneal Topography System and Methods. U.S. Patent #11,890,054. February 6, 2024.

17. Wallace D, Klyce SD. Cloud-based corneal surface difference mapping system and method. US Patent #12,502,067 B2. December 23, 2025.

WHITE PAPERS/NON-PEER REVIEWED ARTICLES

1. Klyce SD, Barnebey HS, Brown T, Habash R, Lyons C, Introduction to tele-ophthalmology. a guide for ophthalmologists. ASCRS•ASOA Telemedicine Task Force. 2018.

2. McDonald MB, Holladay JT, Klyce SD, Alpins NA, Krueger RR: The future of wavefront refraction as a diagnostic tool. EyeWorld 2000;5(5).

3. McDonald MB, Koch DD, Lane SS, Naranjo-Tackman R, Klyce SD, Sher NA, Steinert RF: Preventing and managing unhappy refractive patients. EyeWorld 2000;5(6).

4. McDonald MB, Moderator, with panelists Dausch D, Ditzen KN, Hardten DR, Kezirian GM, Klyce SD, Maus MJ, Mrochen M, Neuhann TF, Pettit G, Tamayo GE: The future of wavefront refractions and customized ablations. EyeWorld 2000;5(10):40-44.

5. Klyce SD, McDonald MB: The future of corneal surgery. *Cat. Ref. Surg. Today* Jan 2024 24(1):1.

6. Klyce SD, McDonald MB: A new approach to thermal keratoplasty. Will it finally work? *Cat. Ref. Surg. Today* Jan 2024 24(1):51-52,64.

7. Klyce SD, McDonald MB, Trattler W: Refractive. Irregular astigmatism: what we're missing— and how to find it. *EYEWORLD*. Spring 2026 pp 70-73. https://digital.eyeworld.org/i/1543566-spring-

2026/72?

WEBINARS

1. Klyce SD, Habash R. *Implementing Tele-Ophthalmology During the COVID-19 Pandemic*. March 26, 2020.

2. Klyce SD, Habash R. *Update: Implementing Tele-Ophthalmology During the COVID-19 Pandemic*. April 9, 2020.

PEER REVIEWED ARTICLES

1. Mishima S, Gasset A, Klyce SD, Baum J: Determination of tear volume and tear flow. *Invest Ophthalmol* 5:264-276, 1966

2. Stanley JA, Mishima S, Klyce SD: In vivo determination of endothelial permeability to water. *Invest Ophthalmol* 5:371-377, 1966

3. Tolpin DW, Klyce SD, Dohlman CH: Swelling properties of dogfish cornea. *Exp Eye Res* 8:429-437, 1969

4. Klyce SD, Dohlman CH, Tolpin DW: In vivo determination of corneal swelling pressure. *Exp Eye Res* 11:220-229, 1971

5. Klyce SD: Electrical profiles in the corneal epithelium. *J Physiol* 226:407-429, 1972

6. Klyce SD, Neufeld AH, Zadunaisky JA: The activation of chloride transport by epinephrine and Db cyclic-AMP in the cornea of the rabbit. *Invest Ophthalmol* 12:127-139, 1973

7. Klyce SD: Relationship of epithelial membrane potentials to corneal potential. Transport processes in the eye. ICER Symposium. *Exp Eye Res* 15:567-575, 1973

8. Klyce SD: Transport of Na, Cl and water by the rabbit corneal epithelium at resting potential. *Am J Physiol* 228:1446-1452, 1975

9. Klyce SD, Maurice DM: Automatic recording of corneal thickness in vitro. *Invest Ophthalmol* 15:550-553, 1976

10. Klyce SD, Wong RKS: Site and mode of adrenaline action on Cl transport across the rabbit corneal epithelium. *J Physiol* 266:777-799, 1977

11. Burstein NL, Klyce SD: Electrophysiological and morphological effects of ophthalmic preparations on the rabbit corneal epithelium. *Invest Ophthalmol Vis Sci* 16:899-911, 1977

12. Klyce SD: Enhancing fluid secretion by the corneal epithelium. *Invest Ophthalmol Vis Sci* 16:968-973, 1977

13. Klyce SD, Russell SR: System for monitoring the thickness of transparent layered structures. *Rev Sci Instrum* 49:1318-1321, 1978

14. Beuerman RW, Klyce SD: A practical solution to low frequency FM tape recording. *Medical and Biological Engineering and Computing* 17:138-140, 1979

15. Klyce SD, Russell SR: Numerical solution of coupled transport equations applied to corneal hydration dynamics. *J Physiol* 292:107-134, 1979

16. Lambert SR, Klyce SD: The origins of Sattler's veil. *Am J Ophthalmol* 91:51-56, 1981

17. Klyce SD: A temperature controller for electrophysiological experiments. *J Electrophys Techniques* 7(2):5-8, 1981

18. Werblin TP, Klyce SD: Epikeratophakia: the surgical correction of aphakia I. Lathing of corneal tissue. *Curr Eye Res* 1:123-129, 1981

19. Marshall WS, Klyce SD: Cell finder speeds impalements with microelectrodes. *Pflugers Archiv* 391:258-259, 1981

20. Klyce SD: Stromal lactate accumulation can account for corneal edema osmotically following epithelial hypoxia in the rabbit. *J Physiol* 321:49-64, 1981

21. Berkowitz RA, Klyce SD, Salisbury JD, Kaufman HE: Fluorophotometric determination of the corneal epithelial barrier after penetrating keratoplasty. *Am J Ophthalmol* 92:332-335, 1981

22. Klyce SD, Palkama KA, Neufeld AH, Harkonen M, Marshall WS, Huhtaniitty S, Mann KP: Neural serotonin stimulates chloride transport in the rabbit corneal epithelium. *Invest Ophthalmol Vis Sci* 23: 181-192, 1982

23. Neufeld AH, Ledgard SE, Jumblatt MM, Klyce SD: Serotonin-stimulated cyclic AMP synthesis in the rabbit corneal epithelium. *Invest Ophthalmol Vis Sci* 23:193-198, 1982

24. Klyce SD, Marshall WS: Effect of Ag+ on the electrophysiology of the rabbit corneal epithelium. *J Membrane Biol* 66:133-144, 1982

25. Werblin TP, Klyce SD: Epikeratophakia: The surgical correction of myopia. I. Lathing of corneal tissue. *Curr Eye Res* 1:591-597, 1982

26. Marshall WS, Klyce SD: Numerical procedures for ion-sensitive electrodes. *J Electrophysiol Techniques* 9:113-121, 1982

27. Marshall WS, Klyce SD: Cellular and paracellular pathway resistances in the "tight" Cl^--secreting epithelium of the rabbit cornea. *J Membrane Biol* 73:275-282, 1983

28. Safir A, McDonald MB, Klyce SD, Werblin TP, Kaufman HE: The cornea press: restoring donor corneas to normal dimensions and hydration before cryolathing. *Ophthalmic Surg* 14:327-331, 1983

29. Klyce SD, Rozsa A: RIPS: a UNIX-based information retrieval program for scientists. *Int J Bio-Med Comput* 14:389-401, 1983

30. Berkowitz RA, Klyce SD, Kaufman HE: Aqueous hyposecretion after penetrating keratoplasty. *Ophthalmic Surg* 15:323-324, 1984

31. Crosson CE, Beuerman RW, Klyce SD: Dopamine modulation of active ion transport in rabbit corneal epithelium. *Invest Ophthalmol Vis Sci* 25:1240-1245, 1984

32. Marshall WS, Klyce SD: Cellular mode of serotonin action on Cl^- transport in the rabbit corneal epithelium. *Biochim Biophys Acta* 778:139-143, 1984

33. Klyce SD: Computer-assisted corneal topography: high resolution graphical presentation and analysis of keratoscopy. *Invest Ophthalmol Vis Sci* 25: 1426-1435, 1984

34. Klyce SD, Beuerman RW, Crosson CE: Alteration of corneal epithelial transport by sympathectomy. *Invest Ophthalmol Vis Sci* 26:434-442, 1985

35. Klyce SD, Crosson CE: Transport processes across the rabbit corneal epithelium: A review. *Curr Eye Res* 4:323-331, 1985

36. Klyce SD, Beuerman RW: The effects of contact lenses on the normal physiology and anatomy of the cornea: Symposium Summary. *Curr Eye Res* 4:719-744, 1985

37. Klyce SD, Ochsner RD: Method for quantitative evaluation of corneal shape changes by contact lens molding. In, The effects of contact lenses on the normal physiology and anatomy of the cornea: Symposium Summary. *Curr Eye Res* 4:727-729, 1985

38. Suarez E, Arffa RC, Salmeron B, Verity SM, Klyce SD, McDonald MB, Kaufman, HE: Efficacy of surgical modifications in myopic epikeratophakia. *J Refract Surg* 4:156-159, 1985

39. McDonald MB, Klyce SD, Suarez H, Kandarakis A, Friedlander MH, Kaufman HE: Epikeratophakia for myopia correction. *Ophthalmology* 92:1417-1422, 1985

40. Arffa RC, Klyce SD, Busin M: Keratometry in epikeratophakia. *J Refract Surg* 2:61-64, 1986

41. Crosson CE, Klyce SD, Beuerman RW: Epithelial wound closure in the rabbit cornea: A biphasic process. *Invest Ophthalmol Vis Sci* 27:464-473, 1986

42. Crosson CE, Klyce SD, Bazan HEP, Bazan NG: The effect of phorbol esters on the chloride secreting epithelium of the rabbit cornea. *Curr Eye Res* 5:535-541, 1986

43. Maguire LJ, Singer DE, Klyce SD: Graphic presentation of computer - analyzed keratoscope photographs. *Arch Ophthalmol* 105:223-230, 1987

44. Maguire LJ, Klyce SD, Singer DE, McDonald MB, Kaufman HE: Corneal topography in myopic patients undergoing epikeratophakia. *Am J Ophthalmol* 103:404-416, 1987

45. Maguire LJ, Klyce SD, Sawelson H, McDonald MB, Kaufman HE: Visual distortion after myopic keratomileusis. Computer analysis of keratoscope photographs. *Ophthalmic Surg* 18:352-356, 1987

46. Maguire LJ, Klyce SD, McDonald MB, Kaufman HE: Corneal topography of pellucid marginal degeneration. *Ophthalmology* 94:519-524, 1987

47. Klyce SD: Corneal modeling system: Common standards. *Shape* 1:1-3, 1987

48. Dingeldein SA, Klyce SD: Imaging of the cornea. *Cornea* 7:170-182, 1988

49. Dingeldein SA, Klyce SD: The topography of normal corneas. *Arch Ophthalmol* 107:512-518, 1989

50. Limberg MB, Dingeldein SA, Green MT, Klyce SD, Insler MS, Kaufman HE: Corneal compression sutures for the reduction of astigmatism after penetrating keratoplasty, *Am J Ophthalmol*, 108:36-42, 1989

51. Bonanno JA, Klyce SD, Cragoe EJ: Mechanism of chloride uptake in rabbit corneal epithelium. *Am J Physiol* 257 (*Cell Physiol* 26):C290-C296, 1989

52. Uusitalo RJ, Lehtosalo J, Klyce SD: One-year follow-up of epikeratophakia for keratoconus, *Graefes Arch Clin Exp Ophthalmol* 227:401-407, 1989

53. Dingeldein SA, Klyce SD, Wilson SE: Quantitative descriptors of corneal shape derived from computer-assisted analysis of photokeratographs. *Refractive & Corneal Surgery* 5:372-378, 1989

54. Wang J, Rice DA, Klyce SD: A new reconstruction algorithm for improvement of corneal topographical analysis. *Refractive & Corneal Surgery* 5:379-387, 1989

55. Klyce SD, Wilson SE: Methods of analysis of corneal topography. *Refractive & Corneal Surgery* 5:368-371, 1989

56. McDonald MB, Frantz JM, Klyce SD, Salmeron B, Beuerman RW, Munnerlyn CR, Clapham TN, Koons SJ, Kaufman HE: One year refractive results of central photorefractive keratectomy for myopia in the nonhuman primate cornea. *Arch Ophthalmol* 108:40-47, 1990

57. McDonald MB, Frantz JM, Klyce SD, Beuerman RW, Varnell R, Munnerlyn CR, Clapham TN, Salmeron B, Kaufman HE: Central photorefractive keratectomy for myopia. The blind eye study. *Arch Ophthalmol* 108:799-808, 1990

58. Kwok LS, Klyce SD: Theoretical basis for an anomalous temperature coefficient in swelling pressure of rabbit corneal stroma. *Biophys J* 57:657-662, 1990

59. Wilson SE, Lin DTC, Klyce SD, Insler MS: Terrien's marginal degeneration: corneal topography. *Refractive & Corneal Surgery* 6:15-20, 1990

60. Lin DTC, Wilson SE, Klyce SD, Insler MS, Kaufman HE: Topographic changes that occur with 10-0 suture removal following penetrating keratoplasty. *Refractive & Corneal Surgery* 6:21-25, 1990

61. Reidy JJ, McDonald MB, Klyce SD: The corneal topography of epikeratophakia. *Refractive & Corneal Surgery* 6:26-31, 1990

62. Wilson SE, Lin DTC, Klyce SD, Reidy JJ, Insler MS: Topographic changes in contact lens-induced corneal warpage. *Ophthalmology* 97(6):734-744, 1990

63. Wilson SE, Lin DTC, Klyce SD, Reidy JJ, Insler MS: Rigid contact lens decentration-a risk factor for corneal warpage. *CLAO J* 16:177-182, 1990

64. Hamano H, Sawa M, Maeda N, Hara Y, Iwasaki N, Tanaka F, Hamano T, Hamano K, Klyce SD, Sakamoto R, Mitsunaga S, Kotani S: Fundamental and clinical studies of corneal physiology and contact lenses. Part I. *Asia-Pacific Journal of Ophthalmology* 2:42-52, 1990

65. Lin DTC, Wilson SE, Reidy JJ, Klyce SD, McDonald MB, Kaufman HE, McNeill JI: An adjustable single running suture technique to reduce postkeratoplasty astigmatism: A preliminary report. *Ophthalmology* 97:934-938, 1990

66. Uusitalo RJ, Lehtosalo J, Klyce SD: Two year follow-up of epikeratophakia for keratoconus. *Acta Ophthalmologica* 68:53-61, 1990

67. Hamano H, Sawa M, Maeda N, Hara Y, Iwasaki N, Tanaka F, Hamano T, Hamano K, Klyce SD, Sakamoto R, Mitsunaga S, Kotani S, Watanabe K: Fundamental and clinical studies of corneal physiology and contact lenses. Part II. *Asia-Pacific Journal of Ophthalmology* 2:108-111, 1990

68. Wilson SE, Klyce SD: Advances in the analysis of corneal topography. *Surv Ophthalmol* 35:269-277, 1991

69. Wang J, Rice DA, Klyce SD: Analysis of the effects of astigmatism and misalignment on corneal surface reconstruction from photokeratoscopic data. *Refractive & Corneal Surgery* 7:129-140, 1991

70. Hamano H, Sawa M, Maeda N, Hara Y, Iwasaki N, Tanaka F, Hamano T, Hamano K, Klyce SD, Sakamoto R, Mitsunaga S, Kotani S, Watanabe K: Fundamental and clinical studies of corneal physiology and contact lenses. Part III. *Asia-Pacific Journal of Ophthalmology* 3:37-41, 1991

71. Wilson SE, Lin DT, Klyce SD: The corneal topography of keratoconus. *Cornea* 10:2-8, 1991

72. Wilson SE, Klyce SD: Quantitative descriptors of corneal topography: A clinical study, *Arch Ophthalmol* 109:349-353, 1991

73. Hamano H, Sawa M, Maeda N, Hara Y, Iwasaki N, Tanaka F, Hamano T, Hamano K, Klyce SD, Sakamoto R, Mitsunaga S, Kotani S, Watanabe K: Fundamental and clinical studies of corneal physiology and contact lenses. Part IV. *Asia-Pacific Journal of Ophthalmology* 3:77-82, 1991

74. Wilson SE, Klyce SD, McDonald MB, Liu JC, Kaufman HE: Changes in corneal topography after excimer laser photorefractive keratectomy for myopia, *Ophthalmology* 98:1338-1347, 1991

75. McDonald MB, Liu JC, Byrd TJ, AbdelMegeed M, Andrade, HA, Klyce SD, Varnell R, Munnerlyn CR, Clapham T, Kaufman HE: Central photorefractive keratectomy for myopia: partially sighted and normally sighted eyes. *Ophthalmology* 98:1327-1337, 1991

76. Kwok LS, Klyce SD: Physiological effects of tert-butyl hydroperoxide on the rabbit corneal epithelium. *CLAO J* 18:97-100, 1992

77. Wilson SE, Friedman RS, Klyce SD: Contact lens manipulation of corneal topography after penetrating keratoplasty: A preliminary study. *CLAO J* 18:177-182, 1992

78. Ruiz-Montenegro J, Mafra CH, Wilson SE, Jumper JM, Klyce SD, Mendelson EN: Corneal topographic alterations in normal contact lens wearers. *Ophthalmology* 100:128-134, 1993

79. Salmeron B, McDonald M, Beuerman R, Klyce S, Verity S, Hanson J, Kaufman HE, Quintana M: Estudio experimental sobre epiqueratofaquia. Revista de la Sociedad Espanola de Oftalmologia, in press

80. Klyce SD, Smolek MK: Corneal topography of the excimer laser photorefractive keratectomy. *J Cataract Refract Surg* 19(suppl):122-130, 1993.

81. Wilson SE, Klyce SD, Husseini ZM: Standardized color-coded maps for corneal topography. *Ophthalmology* 100:1723-1727, 1993.

82. Droutsas D, Miltsakakis D, Tsioulias G, Klyce S: Corneal topography. *Greek Annals Ophthalmol* 30:39-48, 1993.

83. Wilson SE, Klyce SD: Screening for corneal topographic abnormalities prior to refractive surgery. *Ophthalmology* 101:147-152, 1994.

84. Maeda N, Klyce SD, Hamano H: Alteration of corneal asphericity in RGP contact lens-induced warpage. *CLAO J* 20:27-31, 1994.

85. Maeda N, Klyce SD, Smolek MK, Thompson HW: Automated keratoconus screening with corneal topography analysis. *Invest Ophthalmol Vis Sci* 35:2749-2757, 1994.

86. Maeda N, Klyce SD: Videokeratography in contact lens practice. *Int Contact Lens Clinic* 21:1-7, 1994

87. Smolek MK, Klyce SD, Maeda N: Keratoconus and corneal warpage analysis using the keratomorphic diagram. *Invest Ophthalmol Vis Sci* 35:4192-4204, 1994.

88. Maeda N, Klyce SD, Smolek MK: Application of neural networks to the classification of corneal topography: preliminary demonstration. *Invest Ophthalmol Vis Sci* 36, 1327-1335, 1995.

89. Maeda N, Klyce SD, Smolek MK: Comparison of methods for detecting keratoconus using videokeratography. *Arch Ophthalmol* 113:870-874, 1995, and *Arch Ophthalmol [Edición Española]* 6:319-324, 1995.

90. Mafra CH, Dave AS, Pilai CT, Klyce SD, Wilson SE: Prospective study of corneal topographic changes produced by extracapsular cataract surgery. *Cornea* 15:196-203, 1996.

91. Alimisi S, Miltsakakis D, Klyce S: Corneal topography for intraocular lens calculations. *J Refract Surg* 12:S309-S311, 1996.

92. Martinez CE, Klyce SD: Corneal topography in cataract surgery. *Curr Opinion Ophthalmol* 7:31-38, 1996.

93. Kalin NS, Maeda N, Klyce SD, Hargrave S, Wilson SE: Automated topographic screening for keratoconus in refractive surgery candidates. *CLAO J* 22:164-167, 1996.

94. McQueen BR, Martinez CE, Klyce SD: Corneal topography in cataract surgery. *Curr Opinion Ophthalmol* 8:22-28, 1997.

95. Bucci FA Jr, Evans RE, Moody KJ, Tanner JB, Capozza RC, Klyce SD: The annular tinted contact lens syndrome: Corneal topographic analysis of ring-shaped irregular astigmatism caused by annular tinted contact lenses. *CLAO J* 23:161-167, 1997.

96. Maeda N, Klyce SD, Smolek MK, McDonald MB: Disparity of keratometry readings and corneal power within the pupil after refractive surgery for myopia. *Cornea* 16:517-524, 1997.

97. Maeda N, Klyce SD, Smolek MK, Hamano H, Mitsunaga S, Watanabe K: Videokeratography for quantitative surface analysis of used contact lenses. Japan J Ophthalmol 41:235-239, 1997.

98. Smolek MK, Klyce SD: Current keratoconus detection methods compared with a neural network approach. *Invest Ophthalmol Vis Sci* 38:2290-2299, 1997.

99. Smolek MK, Klyce SD: Cinemakeratography using computer morphing. *Optom VisSci*, 74:970-5, 1997.

100. Holland DR, Maeda N, Hannush SB, Riveroll LH, Green MT, Klyce SD, Wilson SE: Unilateral keratoconus. Incidence and quantitative topographic analysis. *Ophthalmology* 104:1409-1413, 1997.

101. Martinez CE, Applegate RA, Klyce SD, McDonald MB, Medina JP, Howland HC: Effect of pupillary dilation on corneal optical aberrations after photorefractive keratectomy. *Arch Ophthalmol*. 116:1053-1062, 1998.

102. Maeda N, Klyce SD, Tano Y: Detection and classification of mild irregular astigmatism in patients with good visual acuity. *Surv Ophthalmol* 43:53-58, 1998.

103. Smolek MK, Oshika T, Klyce SD, Maeda N, Haight DH, McDonald MB: Topographic assessment of irregular astigmatism after photorefractive keratectomy. *J Cataract Refract Surg* 24:1079-1086, 1998.

104. Oshika T, Klyce SD: Corneal topography in laser in situ keratomileusis (LASIK). *Seminars in Ophthalmol* 13:64-70, 1998

105. Oshika T, Klyce SD, Smolek MK, McDonald MB: Corneal hydration and central islands after excimer laser photorefractive keratectomy. *J Cataract Refract Surg*, 24:1575-1580, 1998.

106. Ribeiro JEC, McDonald MB, Klyce SD. Excimer laser photorefractive keratectomy resulting in more myopia postoperatively than preoperatively. Arq. bras. oftalmol 1998;61:181-2.

107. Oshika T, Klyce SD, Applegate RA, Howland HC, El Danasoury MA: Comparison of corneal wavefront aberrations after photorefractive keratectomy and laser in situ keratomileusis, *Am J Ophthalmol* 127:1-7, 1999.

108. El Danasoury, MA, El Maghraby A, Klyce SD, Mehrez K: Comparison of photorefractive keratectomy with excimer laser in situ keratomileusis in correcting low myopia (from -2.00 to -5.50 Diopters. A randomized study. *Ophthalmology* 106:411-420, 1999.

109. Coorpender SJ, Klyce, SD, McDonald, MB, Doubrava, MW, Kim C, Tan AL, Srivannaboon S: Corneal topography of small beam tracking excimer laser photorefractive keratectomy. *J Cataract Refract Surg* 25:675-684, 1999.

110. El-Maghraby A, Salah T, Waring GO III, Klyce S, Ibrahim O: Randomized bilateral comparison of excimer laser in situ keratomileusis and photorefractive keratectomy for 2.50 to 8.00 D. *Ophthalmology* 106:447-457, 1999.

111. Endl MJ, Martínez CE, Klyce SD, McDonald MB, Coorpender SJ, Applegate RA, Howland HC: Irregular astigmatism after photorefractive keratectomy. *J Refract Surg* 15: S249-51, 1999.

112. Oshika T, Klyce SD, Applegate RA, Howland HC: Changes in corneal wavefront aberrations with aging. *Invest Ophthalmol Vis Sci* 40:1351-1355, 1999.

113. Kemp JR, Martinez CE, Klyce SD, Coorpender SJ, McDonald MB, Lucci L, Lynn MJ, Waring GO III: Diurnal fluctuations in corneal topography 10 years after radial keratotomy in the Prospective Evaluation of Radial Keratotomy. *J Cataract Refract Surg* 25:904-910, 1999.

114. El-Maghraby A, Salah T, Waring GO, Klyce S, Ibrahim O. Lasik preferable to prk for −2.5 to −8.00 myopia. *Evidence-Based Eye Care* 1:10-11, 1999.

115. Ruberti JW, Klyce SD, Smolek MK, Karon MD: Anomalous acute inflammatory response in rabbit corneal stroma. *Invest Ophthalmol Vis Sci* 41:2523-2530, 2000.

116. Klyce SD: Corneal topography and the new wave. *Cornea* 19:723-729, 2000.

117. Smolek MK, Klyce SD. Is keratoconus a true ectasia? An evaluation of corneal surface area. *Arch Ophthalmol* 118: 1179-1186, 2000.

118. Endl MJ, Klyce SD: What's new in corneal topography? *Comp Ophthalmol Update* 1:359-364, 2000.

119. Goto T, Klyce SD, Zheng X, Maeda N, Kuroda T, Ide C: Gender and age related differences in corneal topography, *Cornea* 20:270-276, 2001.

120. Endl MJ, Martínez CE, Klyce SD, McDonald MB, Coorpender SJ, Applegate RA, Howland HC: Effect of larger ablation zone and transition zone on corneal optical aberrations after PRK, *Arch Ophthalmol* 2001;119,1159-1164

121. Smolek MK, Klyce SD: Screening of prior refractive surgery by a wavelet-based neural network. *J Cataract Refract Surg*. 2001;27:1926-31.

122. Klyce SD: Developments in corneal topographic analysis following contact lens wear and refractive surgery. *Contact Lens & Anterior Eye* 2001; 24:168-174.

123. Smolek MK, Klyce SD, Hovis JK: The universal standard scale: proposed improvements to the ANSI standard corneal topography map, *Ophthalmology* 2002;109:361-9.

124. Smolek MK, Klyce SD, Sarver EJ: Inattention to nonsuperimposible midline symmetry causes wavefront analysis error. *Arch Ophthalmol* 2002;120:439-447.

125. Klyce SD, Endl MJ: Corneal topography in modern refractive surgery. *Int Ophthalmol Clin.* 2002; 42:19-30.

126. Ambrósio R, Klyce SD, Wilson SE: Corneal topographic and pachymetric screening of refractive surgery patients. *J Refractive Surg* 2003;19:24-29.

127. Ruberti JW, Klyce SD: NaCl osmotic perturbation can modulate hydration control in rabbit cornea. *Exp Eye Res*. 2003; 76:349-359.

128. Goto T, Zheng X, Klyce SD, Kataoka H, Uno T, Karon M, Tatematsu Y, Bessyo T, Tsubota K, Ohashi Y: A new method for tear film stability analysis using videokeratography. *Am J Ophthalmol.* 2003; 135:607-612.

129. Karon MD, Klyce SD: Effect of inhibition of inflammatory mediators on trauma-induced stromal edema, *Invest Ophthalmol Vis Sci* 2003;44:2507-2511.

130. Smolek MK, Klyce SD: Zernike polynomials are inadequate to represent higher order aberrations in the eye, *Invest Ophthalmol Vis Sci* 2003;44:4676-4681.

131. Goto T, Zheng X, Klyce SD, Kataoka H, Uno T, Yamaguchi M, Karon M, Hirano S, Okamoto S, Ohashi Y: Evaluation of the tear film stability after laser in situ keratomileusis using the tear film stability analysis system. *Am J Ophthalmol*. 2004;137:116-20.

132. Courville CB, Smolek MK, Klyce SD: Contribution of the ocular surface to visual optics. *Exptl Eye Res* 2004;78:417-25.

133. Klyce SD, Karon MD, Smolek MK: Advantages and disadvantages of the zernike expansion for representing wave aberration of the normal and aberrated eye. *J Refractive Surg* 2004;20:S537-S541.

134. Eiferman, RA, Klyce S, MacRae S, Maloney RK, Wilson SE: Finding the best wavefront fit. *Rev. Refractive Surg.* 2004; 5:27-32.

135. Smolek MK, Klyce SD: Goodness-of-prediction of zernike polynomial fitting to corneal surfaces. *J Cataract Refract Surg* 2005;31:2350-5.

136. Klyce SD, Karon MD, Smolek MK: Screening patients with the corneal navigator. *J Refractive Surg*. 2005;21:S617-22.

137. Smolek MK, Klyce SD: Absolute color scale for improved diagnostics with wavefront error mapping. *Ophthalmology* 2007;114:2022-30.

138. Tabernero J, Klyce SD, Sarver EJ, Artal P: Functional Optical Zone of the cornea. *Invest Ophthalmol Vis Sci* 2007;48:1053–60.

139. Waring GO IV, Klyce SD. Corneal Inlays for the Treatment of Presbyopia. *International Ophthalmology Clinics* 2011;51:51-62.

140. Ramos IC, Correa R, Guerra FP, Trattler W, Belin MW, Klyce SD, Fontes BM, Schor P, Smolek MK, Dawson DG, Chalita MR, Cazal JO, Ruiz M, Randleman JB, Ambrósio R Jr. Variability of subjective classifications of corneal topography maps from LASIK candidates. *J Refract Surg.* 2013;29:770-5.

141. Packer M, Klyce SD, Smith C. The LENSAR® Laser System–fs 3D for Femtosecond Cataract Surgery. *European Ophthalmic Review*, 2014;8(2):93–8.

142. Packer M, Klyce SD, Smith C. The LENSAR® Laser System–fs 3D for Femtosecond Cataract Surgery. *US Ophthalmic Review,* 2014;7(2):89–94.

143. Randleman JB, Akhtar J, Lynn MJ, Ambrósio R Jr, Dupps WJ Jr, Krueger RR, Klyce SD. Comparison of objective and subjective refractive surgery screening parameters between regular and high-resolution Scheimpflug imaging devices. *J Cataract Refract Surg.* 2015; 41:286–294.

144. Klyce SD, McDonald MB, Morales MU. Screening cataract patients with microperimetry for premium IOL candidacy. *J Refract Surg*, 2015;31:690-6.

145. Koh S, Maeda N, Ogawa M, Asonuma S, Takai Y, Maruyama K, Klyce SD, Nishida K. Fourier Analysis of Corneal Irregular Astigmatism Due to the Anterior Corneal Surface in Dry Eye. *Eye Contact Lens*. 2019 May;45(3):188-194.

146. Koh S, Inoue R, Maeda N, Kabata D, Shintani A, Jhanji V, Klyce SD, Maruyama K, Nishida K. Long-term Chronological Changes in Very Asymmetric Keratoconus. *Cornea*, 2019 May;38(5):605-611.

147. Klyce SD. 12. Endothelial pump and barrier function. *Exp Eye Res.* 2020 Jul 11;198:108068. doi: 10.1016/j.exer.2020.108068.

148. McDonald MB, Mychajlyszyn A, Mychajlyszyn D; Klyce SD. Advances in Corneal Surgical and Pharmacological Approaches to the Treatment of Presbyopia. *J Refract Surg*, 2021; 37(6)[Suppl]:S20-S27.

149. Wilson SE, Klyce SD. Detecting Corneal Topographic Disorders Before Surgery. *Corneal Physician*. 2021; Nov. (2):18-22. https://bit.ly/3ENX0OS

150. van den Berg RM, van den Berg A, Rocha KM, de Barros MF, Dodhia M, Shahid M, Klyce SD. Prediction of the small aperture intraocular lens on visual acuity in patients with keratoconus. *J Cataract Refract Surg,* 2024; 50:930–935.

151. van den Berg RM, DeVaro S, Rocha KM, Fetrin de Barros M, Klyce SD. Predicted visual impact of a small aperture IOL in reducing higher order aberrations in post-radial keratotomy patients. *Vision* 2025, 9, 46. https://doi.org/10.3390/vision9020046

152. Barros M, Klyce SD, van den Berg R, Stonecipher K, Al-Mohtaseb Z, Rocha KM. Visual outcomes and depth of focus from ray tracing aberrometry in small aperture optics intraocular lens in eyes with aberrated corneas. Submitted

153. Barros L.S., Batista da Costa Neto A., Esporcatte L.P.G, Salomão M.Q.,de Melo B.F., Amaral D.C., Rezende R.A, Souza M.L., Garcia É.R., Gualdi L., Klyce S.D., MachadoA. & Ambrósio R., Jr, Prevalence and Risk Factors for Keratoconus in Young Adults Assessed with Tomography and Corneal Biomechanics: A Prospective Cross-Sectional Study, *Ophthalmology Science* (2026), doi: https://doi.org/10.1016/j.xops.2026.101109

154. Klyce SD, McDonald M. In Memoriam – Perry Scott Binder. *Cornea*, in press.